

**PERSPECTIVES OF INNOVATION: A STUDY OF HOW INNOVATION IS DEFINED
IN A GROUP OF INDIAN AMERICAN PROFESSIONALS**

by

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Abstract

The definition of innovation varies from individual to individual and between individual and the team or the team and an organization. The perception of innovation and how individuals define innovation can make an organization stay competitive or go out of business. Research had shown that a traditional approach of idea generation as innovation at well intentioned organizations afforded success only 10% of the time after three years. In forming and managing project teams, which are made up of individuals or selling a service or product to a certain demography it is important and beneficial to understand their perception of innovation and if they see innovation as new, change or improvement. This study is built upon the earlier work of Zhuang (1995), Zhuang, Williamson, and Carter (1999), and McLaughlin and (2013). Prior research conducted by Caraballo and McLaughlin (2012) focused on understanding the perception of innovation as defined or understood by a homogeneous demography of Latino American professionals. This dissertation furthers the research conducted by Caraballo and McLaughlin (2012) with a focus on comprehending how a homogenous group Indian American professionals in the United States perceive or define innovation into the three categories of new, improve and change. While innovation may be understood by individuals as a multi-dimensional construct, the research tried to identify the relationship between perceptions, and gender, age and job function for the homogenous group of Indian Americans in the United States. The research followed the approach and methodology used by Caraballo and McLaughlin (2012) and the study is quantitative, exploratory and used web-based surveys to gather responses from respondents.

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CHAPTER 1. INTRODUCTION

Introduction to the Problem

Innovation is a broadly used term with multiple definitions and applies to products, services, processes and business models (Baregheh, Rowley, & Sambrook, 2009). Innovation can be outcomes-based or knowledge-driven and be the outcome of a process or a new process itself (Quintane, Casselman, Reiche, & Nylund, 2011). Thus, articulating a single definition of innovation becomes challenging and restrictive. For an organization to sustain and stay competitive, innovation becomes a critical need (Caraballo & McLaughlin, 2012). Innovation could be the invention of a new product or service or changing an existing process that leads to a suitable and competitive advantage for the organization. As an example, Apple's iPhone and iPod was an innovative product built on an innovative service iTunes and Apps platform (Finkle & Mallin, 2010). The assembly line design for Dell computers, introduced by Michael Dell that allowed customers to build their own computers online as opposed to purchasing a pre-built computer was an innovation in service design. Nissan's approach under the leadership of Carlos Goshn to leverage Reanult's infrastructure to build standardized automobile models, may be seen as diffusion of innovation or adoption of innovation. Even though innovation is crucial to sustenance and continued market competitiveness, due to the various interpretations and definitions, organizations struggle to strategize and overcome the diversified interpretations and understanding between teams and individuals (Bouncken, 2009).

Innovation could be an outcome or an input to a product or a service that an organization delivers, or innovation could be the diffusion or adoption of an innovation that have been used in

another organization, and hence innovation could be incremental, disruptive and/ or sudden (Zhuang, Williamson, & Carter, 1999). McAdam and McClelland (2002) noted that creativity is the precursor (Amabile, 1997) and an integral part of idea generation, which ultimately leads to innovations. Due to the multitude of definitions and review of such literature led Baregheh et al. (2009) to conclude that innovation lacks a clear definition.

Innovation leads to or is a result of some shape or form of change. The delta for change is not fixed and varies to a large and incalculable degree (Baregheh et al., 2009). As delta/ amount of change increases, so does the risk associated with such change varies. The risk could result in personnel changes, friction or sheer newness and navigation in uncharted waters of the market, thereby making the organization and its employees nervous (Baregheh et al., 2009; Birtchnell, 2011;). Risk aversion or risk avoidance and the affinity to take risk are impacted by behaviors imbued in individuals from their respective cultures, which in turn are made from the nationality and region of origin of the individual or location of the organization (Hofstede, 1980; Hofstede, 2001). Cultures differ in beliefs and values which are derived from history and traditions, and hence adding to diversity of the background and intellect that one brings to the table. As a result of cultural diversity, integration of innovation in a specific direction becomes challenging. Cultural diversity on a team brings diverse ideas and increases creativity as part of innovation, although the very same diversity could lead to obstacles within the team and such hindrances, although uncertain, are to be tackled via interaction as every point of friction leads to an innovative opportunity (Bouncken, 2009).

The core of every culture are people. Perceptions, interactions, beliefs and practices of people in a certain given environment along with the history of the area form culture (Hofstede, 1980). Although not a tangible and surmountable aspect, culture is a heavy impact characteristic

as it pertains to perceptions of innovation. Perception of individuals and teams are driven by the culture and experiences of individuals (DeCusatis, 2008). Perceptions are a result of learnings, knowledge and experiences. Hence, perceptions are open for interpretation.

Different generations transfer knowledge in differ ways, using different communication and pragmatic channels. For instance, due to social networks, such as Facebook, Twitter and LinkedIn, generations in the current age leverage social networks to transfer and translate information. In addition information transmitted is across borders and has a wider net for audiences. According to Hofstede (1980), there are three levels to mental programming that vary from universal to individual and collective that would illustrate and help identify collective and individual behaviors. One of the primary drivers for collective behaviors, as noted by Hofstede (1980) is driven via formal, information and experimental learning. For instance, in the current day, Facebook could be a platform for informal and experimental learning. With advancements in technology and proliferation of the internet platform, the gaps between generations and geographies have been minimized. However, different cultures and generation transfer and translate knowledge in different ways, which would also differ in observations, interaction and participation. Thus, when forming a diverse organization, we also receive diverse national cultures, and the precursor to national cultures are perceptions. Reversing the order, it is logical to assume that individual behaviors are influenced by perceptions (Hofstede, 1980) and thus perceptions of innovation vary by culture.

Background of the Study

This research has furthered the work done by Caraballo and McLaughlin (2012) that focused on how Latino American professionals residing in the United States perceived innovation. Caraballo and McLaughlin (2012) built their research on the work done by Zhuang (Zhuang, 1995; Zhuang et al., 1999), which studied and researched the perceptions of managers in small and medium businesses in England as applied to innovation. According to Zhuang et al. (1999), innovation could be an invention of a new product or service, and/ or improvement of an existing product, process or service, and/ or adoption or diffusion of innovation developed elsewhere into an existing service, product or process. Questionnaires were distributed to certain employees of two businesses and data were collected from anonymous responses to the distributed questionnaires. Lack of structured and mechanized or standardized means for idea generation leads to inconsistency and limited transformation from ideas to deeds. As a result, respondents to the Zhuang (1995) and Zhuang et al., (1999) surveys demonstrated unclear understanding on the concept and perceptions of innovations and as to who or how innovation might be fostered in an organization. Hence the lack of clarity on the subject of creativity, although the research found general agreement on the definitions of innovation. Overall responses indicated that creative people existed in all organizations, but the variation in creativity was a result in the diversity of the professions. Hence, all professions were not deemed to be equally creative. The research clearly indicated the lack of a structured means to integrate and assimilate innovation into an organization's DNA in a structured and methodical manner, regardless of the innovation being applied to a product, service or process.

As part of McAdam and McClelland's (2002) literature review, McAdam and McClelland (2002) used the definition of innovation as laid out by Zhuang et al. (1999) and

researched the connection/ difference between idea generation at a team level and individual level. McAdam and McClelland (2002) identified the linkage between creativity and idea generation as creativity being the precursor for innovation. In addition, McAdam and McClelland (2002) listed motivation, skill, and expertise as components of idea generation. Although, McAdam and McClelland (2002) identified the linkage between idea generation and creativity, their research could not ascertain the source of ideas.

The exploratory research conducted by Caraballo and McLaughlin (2012) focused on Latino American demography within the Information Technology (IT) industry. Leveraging the aforementioned research on perceptions of innovation at the individual and team level, Caraballo and McLaughlin (2012) explored the impact of sub-segments such as age, gender and/ or job on perception of innovation at the individual level. The research conducted by Caraballo and McLaughlin (2012) further defined innovation at various levels of the organization, and hence providing greater insight into building cultures of innovation as an outcome and evolution driven model. In conducting the research, Caraballo and McLaughlin (2012), used an internet-based survey tool to gather data from randomly and anonymously self-selected individuals, who completed the surveys.

Caraballo and McLaughlin's (2012) research on innovation perceptions in the Latino American professionals was part of the larger and global research being conducted by the Project Impact Institute, a not-for-profit organization that conducts innovation research at a global level. (Caraballo & McLaughlin, 2012; Caraballo, Bynum, & McLaughlin, 2011).

In their research, Caraballo and McLaughlin (2012) validated the reliability of the results from the research done by Zhuang (1995) and Zhuang et al. (1999), further confirming the variation amongst individuals of a relatively homogenous group, within the same vertical of

business or similar industry, on perception of innovation. Caraballo and McLaughlin (2012) further examined the analysis of results on perception of innovation by categorizing the demography into categories of age, gender and job. From the results it was evident that the variations on perceptions of innovation were minor between female technical and non-technical respondents. On the other hand significant variations were observed between male technical and non-technical respondents. Hence, the thought process on the perception of innovation between males and females varied from a homogenous perspective when associating and within the data block of technical and non-technical job functions. Association of innovation with new ideas or people, regardless of source, emerged from male technical respondents. From a practical applicability perspective, this is a group that can be construed to associate and further the cultivation of innovation in a product arena. On the other hand non-technical males associated innovation with internal activities and hence from an applicability perspective, this could be a group that would foster innovation in the service or process arena.

The research done by Caraballo and McLaughlin (2012) also included an analysis of perception of innovation as viewed or applied from a generational standpoint. For instance, the results showed that Generation Y, ages 10 to 30, as compared to other generations, comprehend innovation in a much more structured sense of an innovation being more of an invention. From a practical applicability standpoint, a person in Gen Y would be much aligned and acceptable to innovation leading to new products or inventions. On the other hand, the rest of the generations seemed open to innovation in services and processes, in addition to products. The applicability of such research would be seen at an organizational level and at an individual level. At the organizational level, the research helped develop innovation and idea generation teams that align

with the organizations mission and vision at an individual level, the research could drive consumer trends in terms of aligning the products and services from a consumer standpoint.

Innovation by its very nature is driven by creativity and the one hindrance to creativity is restriction. Thus, restricting definition for innovation, has a negative effect on competitive advantage and creativity, which ultimately hinders innovation. Research conducted by Caraballo and McLaughlin (2012) confirmed that vast and distinct variation of perceptions of innovation at the individual level and also found that restricting definitions of innovation arrests creativity and competitive sustenance. Variation and distinction on perception of innovation at an individual level is beneficial and of extreme value to an organization that intends to breed creativity via ideation. The challenge however in managing the process of ideation and innovation, is aligning diverse employees as teams, even though the organizational structure might preclude them from being in the same team or reporting to the same individual (Thompson, 1965). Thereby, making it further challenging for the managers to operate in an area with accountability but no authority and ensuring that the innovation efforts are in fact in line with the missions and vision of the organization. Identifying a leader with strategic foresight and coupling with a tactical approach at the individual and team levels can help catalyze the process of idea creation via multiple communication channels and managing the efforts as a project that bolsters the organizational strategic goals of the organization. The value that project teams bring is that the teams are temporary in nature and can focus on strategic aspects within a restricted period with a fixed deliverable, while returning to the tactical jobs upon completion of deliver in an innovative product, service or process.

The research conducted by Caraballo and McLaughlin's (2012) validates prior research and work, and also lays the roadmap for further exploratory work to be conducted in sub-groups

and in association with the Project Impact Institute to observe and understand perception of innovation at an individual and team level at a global, industry and demographic level. Project Impact (Caraballo et al., 2011) with a team of researchers continues the exploratory work in various countries, but there does not appear to be any research done to analyze and study perception of innovation within a certain ethnic group and specifically the Indian American ethnic group in the United States. This dissertation furthered the research conducted by Caraballo and McLaughlin (2012) by applying their methodology and survey to the Indian American homogenous group in the United States.

Statement of the Problem

Innovation is a critical need for all organizations as it adds value and sustains competitive advantage (Baregheh, Rowley, and Sambrook, 2009, p.1323). Businesses need to innovate to increase performance and competitive advantage (McLaughlin, McLaughlin and Preziosi (2004) and as a primary source of wealth creation (de Waal, Maritz, and Shieh, 2010). Innovation is a concept that extends to technology, products, processes, and services. Innovation begins at the individual level with the recognition of a need (Caraballo, and McLaughlin, 2012).

Involving personnel is a necessity for initiating, developing, and implementing an innovation project for both competitive advantages, as ideas and needs drives innovation. Given its critical function, innovation should be a readily identified concept that has a generally accepted definition. Flight, Allaway, Kim and D'Souza (2011) state that ability to measure how individuals perceive innovation is important for identifying problems related to success. Baregheh, et al. (2009), identified over 60 unique definitions of innovation collected from disciplines such as management, economics, technology, knowledge management, and marketing. Given that innovation is a key ingredient for success, is there a simple but effective

method of understanding innovation from an individual perspective (Zhuang, 1995,1998)? In addition, many definitions of innovation exist at the managerial, technical, and functional level. Finally, is there a difference between individual perceptions of innovation across demographic categorizations of gender, job function, and age (generational cohort)?

Research continues on an international basis on how to recognize innovation. Caraballo and McLaughlin (2012) studied a homogeneous Latino population of those individuals employed within the IT profession. Other groups, cultures, and ethnicities require evaluation to determine if the patterns exhibited are consistent throughout different countries and cultures. Innovation is misunderstood, poorly defined, and lacks an understanding of how culture and the individual play a role in identifying innovation. This dissertation examined Indian professionals to determine if their perceptions of innovation follow the patterns established or whether this group diverged from previous groups surveyed.

Purpose of the Study

Understanding how Indian professionals perceive innovation, and identifying if there is a relationship between these perceptions and gender, age and functional association, formulated the purpose of this study. A prior study of Latino American professionals by Caraballo and McLaughlin (2012) was extended via a quantitative exploratory study. Although there are additional studies and research ongoing in different countries, there weren't any similar work focused on the Indian ethnic group located in the United States.

Rationale

Building on the research conducted by Caraballo and McLaughlin (2012), the rationale for conducting this research rose from the opportunity to expand and further research existing knowledge on the methods, mannerisms and beliefs on how various distinct cultural groups

perceived innovation. The research conducted by Caraballo and McLaughlin (2012) focused on the Latino American professionals. Given that limited research exists in this area and because most of the research done thus far looks into culture and innovation at a team, group, organization and national level. In addition, economic and technological changes have stimulated globalization, and hence the call to question the validity of this theoretical model.

Innovation is key to ensuring sustenance of competitive advantage and financial success, whether such success or sustenance is in regard to a product, service or even at a national level (Arbolino, 2011; Caraballo & McLaughlin 2012). There were many reasons for focusing this research on Indian professionals living in the United States. One of the reasons was that the researcher himself was an Indian origin professional working in the United States. The group was large enough to gain a good sample and small enough in terms of an absolute sample. Secondly, as Project Impact expands globally, cultural sub-groups are not a primary focus for research and the entire practice in India could potentially become a vertical for culture for innovation research with a broad horizontal of industries. According to the 2010 US Census, Indian Americans represent 1% of the total US population and the group has seen a 69.3% increase since 2000 census. The group is growing at a rate much higher than the Hispanic/ Latino population, which comprises of 16.3% of the US population as per 2010 Census. Fourthly, according to the 2010 US census data, Indian American households have the highest household income, with a median household income of \$88, 538 (2009 census), amongst all the ethnic groups in the United States. Hence, even though the absolute numbers are smaller for the group, the population selection demonstrates exponential and significant economic impact, both nationally and internationally.

The revised, and previously validated survey instrument used earlier by Caraballo and McLaughlin (2012) and Project Impact was used for this dissertation as a survey instrument.

Keeping consistent with the previous study by Caraballo and McLaughlin (2012), this dissertation used random sample approach.

Research Questions

Cooper and Schindler's (2007) management research hierarchy was used by the researcher to develop the research questions. The objective of this study from this structure was to study and explore the management question: how is innovation perceived by members of a distinct cultural and homogenous group at an individual level? The hope of the researcher was that the answers would provide insights and better understanding of this subject and improve efforts to innovate and generate innovation, both at a team/ vertical and organizational level.

The research questions were:

1. What are the perceptions of Indian professionals residing in the US toward innovation?
2. What is the relationship between gender, generational cohort, functional association, and one's individual (personal) perception of innovation among Indian professionals?

The survey questions were both forced choice and close-ended using a Likert scale.

Significance of the Study

Generation of additional data that would help better understand how individual perceive innovation, extending the previous work done on this subject, and adding to the limited body of research on this subject, are some of the anticipated outcomes and pertinent significance of this study.

Every culture in this world is unique and so are the challenges and coping mechanisms in respective cultures. This study builds on the research of Hofstede (Hofstede, 1980; Hofstede & Bond, 1988; Hofstede & Minkov, 2010) that accepts that cultures develop based on how distinct groups interpret, understand, and cope with challenges unique to their world. The differences in development, comprehension and coping do become apparent in different behaviors, which the researcher recognizes and hence also recognizes that the culture model cited by Hofstede (1980) is not appropriate to be applied at the individual level.

Definition of Terms

Although there are multiple definitions of innovation, the central theme for innovation describes innovation as a new or unique product or service (Baregheh et al., 2009). Zhuang (1995), Zhuang, Williamson, and Carter (1999), and Caraballo and McLaughlin (2012, p. 554), further clarified the definition of innovation to be an invention or creating something new, an improvement or refinement, and/ or diffusion or adoption of innovation that may have been developed elsewhere. Innovation may be broadly defined into new, improved or change innovation, where new and improved innovations may fall under the category of incremental innovations. Change is more in line with radical innovation. This study does not include any other specific or unique terms. All acronyms are completely spelled out when first used.

Assumptions and Limitations

Assumptions

By extending the research of Caraballo and McLaughlin (2012), and leveraging the revised associated questionnaire as a survey instrument, this quantitative study incorporated and used a random sample approach. The researcher assumed that the validity of the survey instrument that was previously established, would not change, even though the cultural group's

participation in this study were different from the previous group. An assumption the researcher made is that there would be at least 100 complete responses with sufficient density in different genders and multiple professions/ verticals/ organizational layers to allow meaningful interpretation and comparison. The researcher further assumed that respondents would be honest and objective in their responses and would truly represent the population. The researcher ensured that no bias or researcher error was introduced; minimized risks to participants; conducted this study in an ethical and controlled manner to ensure the credibility of the results and maintain privacy, confidentiality and data security.

Limitation

Given the focus of the study and expected number of responses the study had a number of limitations. Because of the focus of the study on one cultural group, and the small sample size, the study itself did not allow any room for portability or transferability of the results to other groups. Further research and additional data analysis would be beneficial in larger groups of Indian-Americans (non-immigrants) and Indian-American (naturally born American-Indian) professionals. A larger group and sample would also lend to comparison and validation of the findings of this study. Secondly, variation in cultural backgrounds and sub-set groups may not provide for sufficient respondents, which ultimately lead to development of conclusions. Thirdly, responses from male and female gender groups may not be substantial to develop meaningful and authoritative conclusions about how Indian females understand innovation compared to Indian males in the United States.

Although a previously validated and reliable survey instrument was used by the researcher, it is possible that the characteristics of the respondents could very well affect the reliability and validity of the instrument. Time constraints on the researcher mandated by the

dissertation process, sampling strategies, and methodical constraints in terms of conducting the research and gathering information, as well as limited experience of the researched with conducting such research may limit the usefulness of the results. Last but not the least, inability to use the results as confirmation of information, coupled with inexact nature of sociological research in a diverse environment may limit the certainty of this research.

Theoretical Framework

All cultures are different from one another. And hence the variation as a result of having diverse culture on an innovation team will also vary at an organizational level, team level and at an individual level. However, there are certain aspects and attributes to people that are shared or common amongst cultures. As an example, a common platform of communication amongst people from multiple national cultures might be Facebook. In either case, even though there are significant differences between cultures, it is not possible to ascertain if one culture finds greater financial success over another and hence innovation needs to be understood at an organizational level to have a common understanding and to overcome cultural barriers. Initial research by Hofstede (1980) and, Hofstede and Bond (1988) lead to the identification of four cultural dimensions: power distance index, individualism vs. collectivism, uncertainty and masculinity versus femininity. The four dimensions, identified how members from each culture gravitated towards, power circles, staying individualists versus integrating into a collective grouping, how averse are acceptable were the individuals and culture to uncertainty and ambiguity and finally how did the genders in each culture vary in their perceptions. Hofstede and Minkov (2010) identified two more cultural dimensions from a global study that highlighted long-term orientation or Confucian Dynamism and indulgence versus restraint.

Baregheh et al. (2009) identified three types of innovation as new, improved and change, and as applied to a product, service or process, while trying to formulate a single definition for innovation that could apply to multiple scenarios. Further, definitions of innovation, do not necessarily identify or elaborate on the process and elaborate on how culture influences the perception of innovation. However, the research adds minimal clarity on how perception of innovations is impacted or influenced by specific cultures or ethnicity.

Integrating the concepts of cultural dimensions as identified by Hofstede (1980) and, Hofstede and Bond (1988), along with types of innovation identified by Baregheh et al. (2009), Caraballo and McLaughlin (2012) furthered and explored the relationship between independent variables of gender, job, and age and the dependent variables of innovation defined as new, improved, and change, using the survey instrument and advancing the research conducted by Zhuang (1995,1998). Although there were minor differences between gender, the independent variable, and perception of innovation, the dependent variable, the research highlighted that there is a variation in the perception of innovation depending on the job (technical versus non-technical) and generational cohort.

Collective behaviors, defined as national and organizational cultures are stimulated by the transference of perceptions as explicit and tacit knowledge via formal, informal and experimental learning (Hofstede, 1980). Pictorially, figure 1, shows the flow of such stimulus reaction.

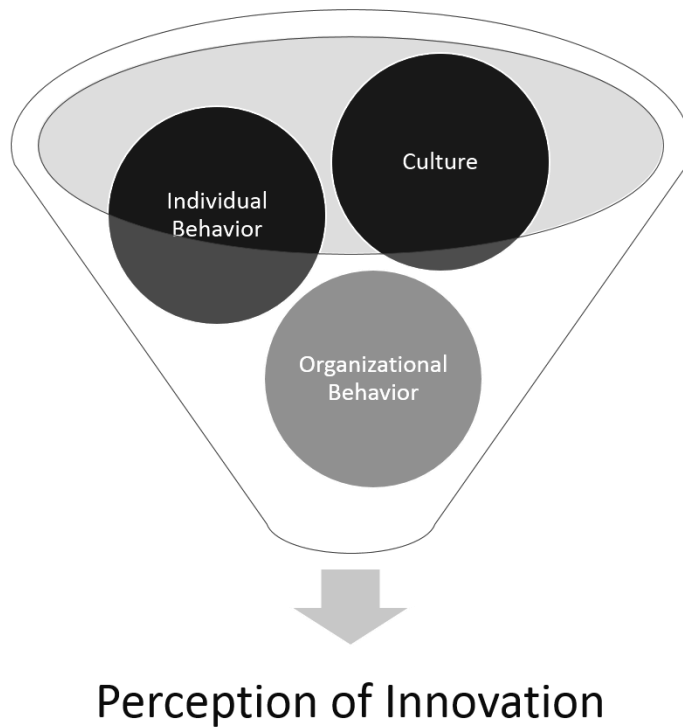


Figure 1. Chain of Stimuli and Influence.

The starting point seems to be perception for comprehension and interpretation of experiences, which varies from individual to individual and culture to culture. The unclear part is as to what is the precursor to perception. Caraballo and McLaughlin (2012) noted that perception of innovation varied when demographics of gender, age and job function were selected. Integrating the concepts, the chain of stimuli and influence may be modified to that shown in Figure 2.

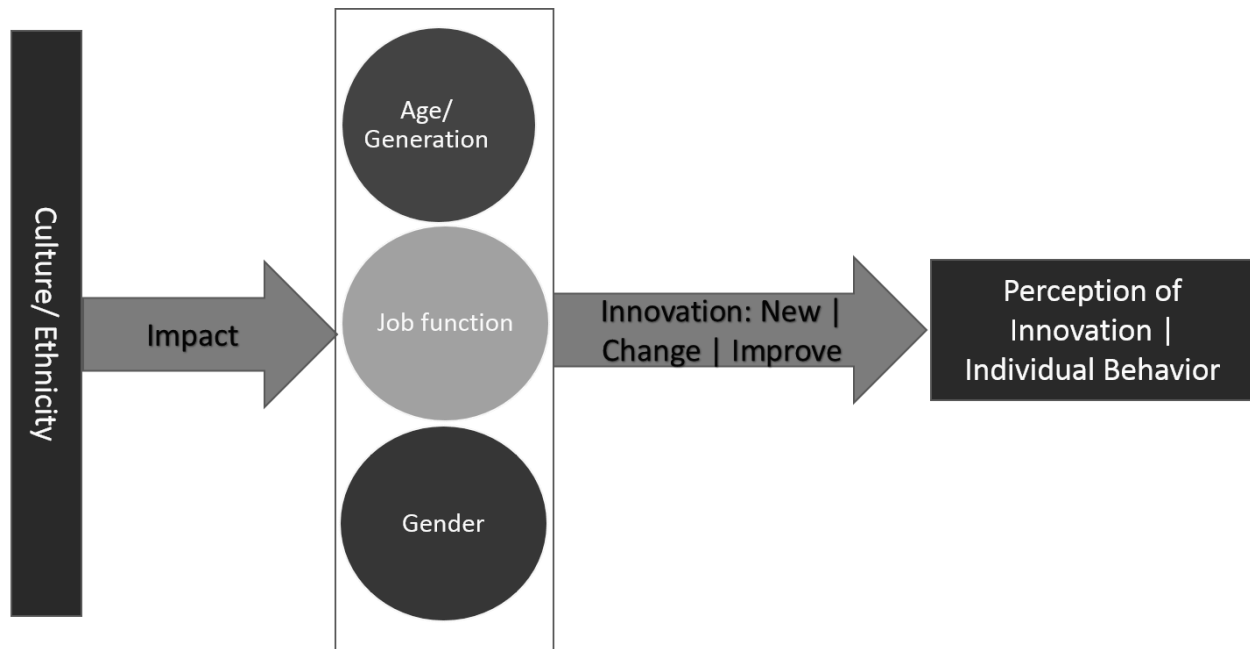


Figure 2. Reduced Chain of Stimuli and Influence.

The theoretical model illustrated by Caraballo's and McLaughlin's (2012) comprising of gender, age, job as independent variables of the selected demography, Indian Americans, and the dependent variable of perception of innovation, whether news, change or improved forms the basis of this study.

Organization of the Remainder of the Study

Chapter 2 of this dissertation includes a literature review, examining the impact on innovation from a cultural perspective as it pertains to nationality, organizational culture, and sustenance of competitive advantage. A detailed description of research along with research design is covered in Chapter 3; the study's findings are presented in Chapter 4, and finally Chapter 5 concludes with interpretations, recommendations and implications as a result and measure of the findings.

CHAPTER 2. LITERATURE REVIEW

Introduction

Innovation is a broad topic and applies to a myriad of scenarios. Hence, articulating one specific definition for innovation that applies in all cases is not possible. Baregheh et al. (2009) identified numerous definitions, approximately 60 definitions over 7 different disciplines, as part of their literature review on the topic of innovation, spanning the time period of 1934 to 2007. Even after such a comprehensive study, Baregheh et al. (2009) provided a general definition for innovation that could potentially be applied in various instances. New, improved and change, were the most frequently use descriptors for innovation and the outcome of innovation was mostly general as opposed being limited or restricted to a process or product or service.

In order to understand perception of innovation among Indian Americans, the context of individual perceptions with an understanding of information, knowledge, and the interconnectivity among individuals to identify with or agree on types of innovation needs to be understood. Throughout this chapter the literature covers the domains of knowledge management with an understanding of knowledge management, and perceptions of how individuals perceive change or improvement and how individuals associate such change with innovation. The Baregheh et al. (2009) research provided various components of innovation, although associating with a general definition, providing the researcher to track a progress further with the research on perception of innovations and afford a platform for analysis of the various components to comprehend as to how culture impacts and influences perceptions of innovation and how outcomes are impacted at a team and an organizational level. Beginning

with idea generation, knowledge management and boundary spanning, both impacted by culture and beliefs, become key section of the value chain that forms the innovation process. The study is an extension of the research conducted by Zhuang (Zhuang, 1995 & Zhuang et al., 1999) and Caraballo and McLaughlin (2012), while applied to the specific segment of Indian-Americans in the United States. Hence, this chapter will introduce the works of by Zhuang (Zhuang, 1995 & Zhuang et al., 1999) and Caraballo and McLaughlin (2012), in an effort to provide the background, context and follow-up with a discussion of the innovation process and the impact of culture on the perceptions of innovation and impact on the process.

Previous Research

Zhuang

In researching the innovation process, Zhuang (Zhuang, 1995; Zhuang et al., 1999) conducted two studies. The first research (Zhuang, 1995) was a pilot study that involved managers in small and medium business in England and the pilot study tested surveys on attitude and activity as it pertained to innovation. According to Zhuang (1995), in a broader and holistic view of the innovation process, there were three components that made up the innovation process: individual creativity, organizational processes/ procedures, and output. Thus, inferring a critical role for the individual as part of the creativity process, and the responsibility of the organization to organize and manage people, process and procedures in an effective manner. After all, creativity only breeds if the individual is allowed to be creative and if the organization's DNA has the appetite and fortitude to experiment and foster creativity.

Innovation has a broad definition as it applies to products, services, processes and procedures (Zhuang, 1995). The definition for innovation formulated by Zhuang (1995) in the broad sense incorporated processes and procedure in the verticals of manufacturing and

management. Zhuang's (1995) comprehension of innovation was that the innovation process is not linear and that the innovation process needed to incorporate change, creativity and innovation. Individuals bring knowledge, skills, and attitudes to the organization and job at hand, thereby making creativity an intangible that is inherent in the individual and made up of the individuals knowledge, skills and attitude (Zhuang, 1995). Amongst knowledge, skills and attitudes, the former two are required and mandated by the job duties and deemed as the minimum requirements for the individual to be in the job that the individual is in. And hence, knowledge and skills can be learnt and trained and may not be unique to the individual, however, attitude is unique to the individual that cannot be repeated. Hence, the relationship of creativity depending on independent attitudes. Zhuang (1995) established the relationship between creativity and attitudes, with the former becoming the dependent variable and the latter being the independent variable.

Although creativity sources from the individual, the innovation process is in fact a collective activity. Zhuang (1995) developed an attitude survey consisting of 28 quantitative questions that focus on identifying the individual, his creativity level from a self-assessment perspective, the individual's comprehension of the concepts of innovation, and identify the factors necessary to maximize the innovation potential of the individual. Zhuang's (1995) study tried to correlate measures of creativity and the organizations achievement of innovative outcomes. In addressing and capturing innovation in the product, process and management areas, Zhuang (1995) developed a 31 question survey that included qualitative and quantitative questions.

The survey results, from both businesses surveyed in the initial study by Zhuang (1995), were inconclusive and contradictory. From the survey results, it looked like males between the

ages of 21 to 30 had a better comprehension of creativity as compared to older males, females and individuals in non-management roles. Although, inconsistency existed in age groups when comparing the two businesses. There was a lack of understanding about innovation in a general and universal sense, but respondents to the surveys did think of innovation from different perspectives of making existing products better, which is included as part of the broader definition of innovation. Answers to questions on the innovation process revealed inconsistency in understanding of responsibilities for creation and implementation (Zhuang, 1995). The results also indicated the need to further refine the survey instrument.

With a modified survey instrument, Zhuang et al. (1999), conducted the second research that studied as to how managers defined innovation and also to identify how managers applied their comprehension of innovation in a pragmatic setting. As part of this study and to set the background, Zhuang et al. (1999), defined innovation as something new, or improvement of an existing product or service, and/ or adoption of an innovation in a new environment or scenario, which would also constitute diffusion of innovation. Zhuang et al. (1999) provided further clarification on the perspective of the innovation process and included input, process, and outcome as part of the innovation process. Zhuang et al. (1999) achieved this change by combining the survey instrument with a corporate modeling survey instrument, although, this did negatively impact the response rate. The respondents in this study were students of the Staffordshire University Business School and the methodology for collection of responses included a questionnaire, which had a 60% response rate, and then followed by an in-depth telephone interview, for which 27 individuals were willing to participate.

Zhuang et al. (1999) applied various statistical techniques such as descriptive statistics, cross-tabulation, ranking, correlational analysis, and selected statistical testing, etc. for analyzing

the results. The demographic break-up of respondents in terms of gender was 51% male and 49% female. In terms of generational cohort or age groups, the respondents fell into three categories: 15% under the age of 25, 35% between the ages of 26 and 35, and 35% over the age of 36. From a managerial versus non-managerial standpoint, the group constituted of 75% managers and 25% non-managers. Respondents came from medium to large enterprises, and from both public and private entities. The industries included manufacturing and service vertical which covered products and services.

The second research by Zhuang et al. (1999) resulted in mixed results that in some cases went against some assumptions. For instance, Zhuang et al. (1999) assumed a positive correlation between creativity and mastery and non-conformity. However, the results showed opposing correlation relative to creativity of engineers relative to mastery and non-conformity. Perception of different job groups was found to be inconsistent while viewing fellow colleagues and creativity was negatively correlated with administrative jobs. Zhuang et al. (1999) noted that middle managers seemed more knowledgeable and had implemented or used some of the innovation techniques as opposed to most respondents who were not even familiar with proper names for the innovation techniques.

Practices that foster an innovative culture and organizational policies, were two aspects researched as part of organizational culture that influenced innovation. Overall the questions targeted policies as they pertained to improvement of products and services, sales and productivity (Zhuang et al., 1999). Two-thirds or 66.7% of the respondents to the surveys responded favorably towards organizational policies towards improving productivity and improvement of products/ services offered by the organization. However, less than 50% of the respondents could identify policies that would increase sales (Zhuang et al., 1999). Although,

the results showed inconsistency, of the 26% of the public sector organizations represented in the sample, 81% of the respondents in this category indicated that they did not have a sales oriented policy, while respondents from the service industry indicated that they had policies for increasing sales or policies that were sales oriented. In fact, the service industry was the industry with more organizations that had a policies for ideation that would encourage sales (Zhuang et al., 1999).

As part of researching organizational culture, respondents were asked to identify, of the 10 statements about creativity and productivity, which were important to them as individuals and which of those were emphasized by their respective organizations (Zhuang et al., 1999). The one item that was most valued by individuals and thought to be most valued by respective organizations was “recognition and appreciation”, which speaks to the social needs of the individual in those organizations. At the individual level, positive correlation was found between mastery and the opportunity for solo work and, mastery and freedom on what to work as it pertained to working on areas of individual’s greatest interest, although non-conformity negatively correlated with opportunity for solo work and freedom on what to work (Zhuang et al., 1999). Contrary to the traditional conception of highly creative people to work in solo, a positive correlation was found between creativity and a stimulating environment, indicating that highly creative individuals fostered a highly stimulating environment. Another positive correlation was between creativity and risk-tolerance. On all of these statements, respondents scored their organizations lower.

Although there was no standard template, the study found that idea generation existed in all organizations (Zhuang et al., 1999). The research conducted by Zhuang et al. (1999) reached a variety of conclusions that differed. The research also found that innovation was driven by dynamic environments despite inconsistency of clear policies in innovation. Hence, even though

clear policies on innovation were not present, dynamic environments were found to be driving innovation (Zhuang et al., 1999). The research found that highly creative employees perceived their organizational cultures to be non-supportive. Finally, the research concluded that managers lacked a clear understanding of idea generation techniques. This further contributed to the negative value associated with creativity training programs (Zhuang et al., 1999).

Both studies, Zhuang (1995) and Zhuang et al. (1999), had agreement on the definition of innovation in general and agreed on the concepts of innovation. However, the results were found to be inconsistent as applied to individual perceptions of creativity and innovation. In addition, a lack of clear understanding on the concept of innovation emerged, in association with the unknown of who was most responsible for the same. Respondents did not perceive that all professions were equally creative, although they believed that creative individuals were present in all organizations. Even though a positive correlation existed between creativity and stimulating environments, creativity training was found to be inconsistently values and techniques to increase idea generation were no well-known. The studies clearly emerged the lack sufficient integration of processes, procedures and structures at an organizational level to support innovation.

Individual creativity composed of knowledge, skills and attitudes, an assumption made in the original study by Zhuang (1995). Knowledge and skills are directly related to the job. These are attributes that one must possess or be trained on in order to perform their job duties or be hired into the job that the individual is in. And hence those attributes may not be unique to the individual. However, attitude is unique to the individual and hence viewed as an independent variable and precursor to creativity as a dependent variable. Further extending, the researcher, Zhuang (1995), viewed creativity as the independent variable to output as the dependent

variable. Juxtaposing, individual attitudes seem to be the independent variables to outcome as the dependent variable. The results identified that younger individuals, aged 21 to 30 years, who were males and, managers appeared to be most creative, although age was not consistent.

Zhuang et al. (1999), the following study to Zhuang (1995) described the innovation process as input, process, and output, with individual creativity as one of several inputs to the innovation process. The assumption made was that mastery and non-conformity were the precursors to creativity and ultimately productivity. The results showed the assumption to be contradictory. Further identifying the inputs, high risk tolerance and a stimulating work environments seemed to be the precursors to creativity. With creativity as the dependent variable, the job group emerged to function as the independent variable. This, paving the transformation-path from individual level to the organizational level. For innovation output as the dependent variable, organizational policies and procedures formed the independent variable as they related to innovation output.

The research and studies conducted by Zhuang (1995) and Zhuang et al. (1999) provided interesting conclusions, especially in regards to creativity at the individual level and also provided conclusions and the template to further explore perception of innovation at the individual and organizational level. Innovation in an organization is driven by a dynamic and stimulating environment, and it is the organization's responsibility to create such an environment. Secondly, even though skills and knowledge are variables between individuals and are precursors to creativity, they tend to be job related. Left alone, or singled out, the attributes do not explain the differences in creativity amongst individuals. Thirdly, individual creativity is a dependent variable and affected by age, gender and job groups, that are independent variables. Last but not the least, the attitude to high risk tolerance environments seemed necessary and a

precursor to increase individuals' creativity. The aforementioned indicate a coalesced and linked role for individual and the organization to identify the precursors to creativity and weave them into the organizations culture, so as to maximize innovation output. The results from these studies proved inconclusive when analyzed using statistical analysis, but the studies provided the grounding to further the research on innovation by providing a framework, model and instrument to carry out further exploratory research.

Caraballo and McLaughlin

The research of Caraballo and McLaughlin (2012) sought to comprehend how perceptions of innovation varied at the individual level within a single, homogeneous cultural group. Starting at the individual level, innovation happens when individuals use creative processes to solve problems. This was the foundation that premised the study by Caraballo and McLaughlin (2012) that was built on previous research (Baregheh et al., 2009; Zhuang, 1995; Zhuang et al., 1999). In performing the research, Caraballo and McLaughlin (2012) leveraged the definition of innovation as something new, improved/ changed or adoption. Diffusion in new scenarios, circumstances or contexts (Zhuang et al., 1999) and Baregheh et al's. (2009) identification that the precursor to the outcome is the perception of innovation at the individual level.

Caraballo and McLaughlin's (2012) research focused on the Information Technology industry vertical and specifically on the Latino Americans working in the Information Technology industry. Caraballo and McLaughlin (2012) collected data using the 10-question survey from Zhuang (1995), which garnered comprehension of innovation at the individual level. Using a 5-point Likert scale, respondents recorded their level of agreement or disagreement. For statistical analysis, Analysis of Variance (ANOVA) technique was used at the 95% confidence

level. In the different meaning of innovation, Caraballo and McLaughlin (2012) identified statistically significant differences, which was different from the Zhuang (1995) study. Caraballo and McLaughlin (2012) noted that some of the differences may have been due to outliers. However, Caraballo and McLaughlin (2012) noted that the Levels statistic of 14.188 supported the conclusion of variation difference for different questions. In order to determine the validity and reliability of the Zhuang (1995) instrument, Caraballo and McLaughlin (2012) performed factor analysis, extending study of the results. Requirements of factor analysis were met by the data, which was normal, linear and had conceptual linkages. Three constructs (independent factors) describing 57% of the total explained variation emerged from the 781 responses (Caraballo & McLaughlin, 2012). The data variation of 57% met the requirement for social scientific research, less than 60%, even though some differences in naming conventions existed.

Caraballo and McLaughlin (2012) performed “Q factor analysis” to examine the demographic characteristics as a precursor of perceptions of innovation. Gender, age group and job function were identified by Caraballo and McLaughlin (2012) as specific demographic variables. Gender and perception of innovation did not carry statistically significant differences between them.

Technical and non-technical respondents and gender delineated job function. Using ANOVA analysis at 95% confidence level, Caraballo and McLaughlin (2012) studied female technical and non-technical respondents perception of innovation. The results did not yield to identify statistically significant differences. However, similar analysis of male respondents using the exact technique produced results that were significantly different. The male (gender) and technical respondents perceived innovation as “identifying the best new ideas and/or best

people” ” (Caraballo & McLaughlin, 2012, p. 559). On the other hand, the male non-technical respondents perceived innovation as related to “attract[ing] the best people or improve those that are available internally” (Caraballo & McLaughlin, 2012, p. 559). The results suggest differences of perception of innovation to be dependent on demographic factors.

Caraballo and McLaughlin (2012) categorized the age groups into three categories: ages between 18 and 30 years old were Generation Y, ages 31 to 44 were Generation X and ages 45 and older were the baby Boomers. With the 10-question survey as the dependent variable and gender, age group and job function as independent variables, Caraballo and McLaughlin (2012) tested a three-way interaction between the dependent and independent variables. This was done prior to testing age as a separate factor. Analysis using the multivariate ANOVA analysis of these variables tested as significant at the 95% confidence level. Caraballo and McLaughlin (2012) noted that different groups can share similar perception, although perceptions of innovation may vary from individual to individual. Caraballo and McLaughlin (2012) identified that older groups seemed to accept all three perceptions of innovation, while respondents in Gen Y were found to have a narrower perception of innovation as improvements or changes being interchangeable with new innovations.

Caraballo and McLaughlin (2012) confirmed and noted that restrictive definitions of innovation could have a negative effect on creativity and competitive advantage, while individual perceptions of innovation vary distinctly. Rather than directing a process, because perceptions of innovation vary at the individual level, organization have a need to manage the innovation process by people and perceptions or in other words aligning the appropriate combination of employees with the required perception. In such a scenario, managing the innovation process might require a tactical approach, both at the individual and team level, that

stimulates, focuses and catalyzes activity that would support the organizations overall strategic plan. If that be the case, the final state may be one of constant realignment and change rather than a static structure.

Caraballo and McLaughlin's (2012) exploratory research was groundbreaking and extremely valuable as the work done provides the transformation roadmap for moving the research forward as well as validating prior works. With the cultural impact on the perception of innovation, Caraballo and McLaughlin (2012) recommended additional research with other distinct groups to further validate the findings, develop and acquire greater responses from other sub-groups to clarify findings and make some modifications to the survey instrument to increase its precision. This dissertation furthers the work of Caraballo and McLaughlin (2012) by applying their survey to a different, homogenous group, Indian Americans in the United States.

Innovation Process

The general definition of innovation is derived from multiple perspectives of the innovation process and applies to multiple scenarios. Thus making a single definition of innovation to gain wide acceptance. The following section further examines creativity, factors that enable knowledge management and boundary spanning and the impact of multiple levels of culture, while maintaining the description of the innovation process to include individual creativity in association with policies, procedures and processes at the organizational level, leading to innovative output (Zhuang et al., 1999).

Stages of Innovation

The innovation process is a multi-stage process, which comprises of at least two steps/stages, generally idea generation and idea implementation, and symbolically and pragmatically characterized as a cyclical process of creation and destruction (Denning, 2012; Kaasa & Vadi,

2008; Leavy, 2011; Roberts, 2007; Williams & McGuire, 2010; Jucevicius, 2010). Baregheh, et al. (2009) noted innovation's definition to be one of a multi stage process.

According to Zhuang (1995), the innovation process is made up of three components: individual creativity, organizational processes and procedures, and output. In the follow-up study, Zhuang et al. (1999), further broadened the definition to include input, process and output. Although a specific definition that covers all aspects is yet to be found, research and studies identify innovation process to include iterative individual and collective actions, being split between idea generation and idea implementation and the pragmatically cyclical creation and destruction. The final product, service or process is the result of the collective actions at the organizational level, which are in turn driven by creativity at the individual level. Transforming ideas to deeds needs resources, time and commitment. Since no organization has an unlimited supply of resources and since resources are limited, only certain ideas that are generated may be selected and found worthy for implementation resulting in a new or improved product service or process. Hence, the need for a screening/ filtering or vetting or prioritization process. In this context, Figure 3 illustrates the basic innovation process.

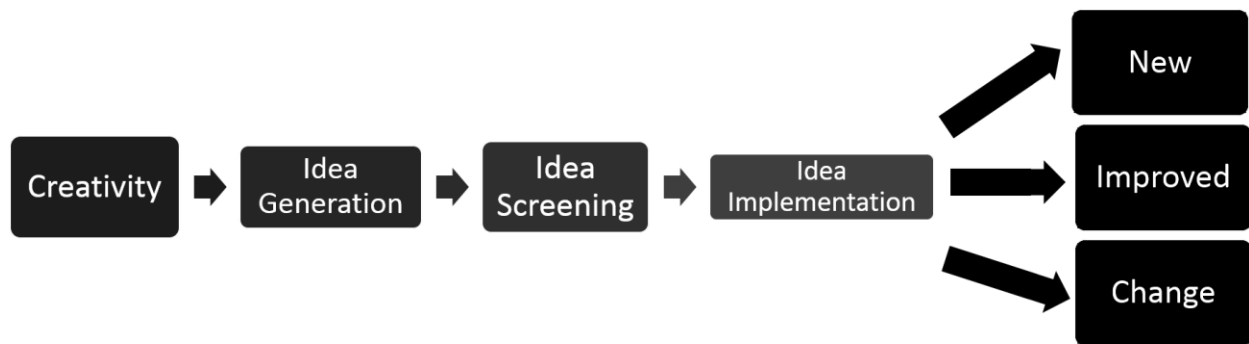


Figure 3. Basic Innovation Process.

The organizations' infrastructure, policies, procedures, structures, perception of innovation and strategy influence across the phases and especially as a feeder and outcome to screening process, which is also influenced by external factors such as the market environment.

Creativity. In the innovation process, the individual and the organization collectively have a role to play and creativity becomes a vital and critical component at the individual level (Zhuang, 1995). Collectively, creativity at the employee's individual level and the degree of interactions amongst employees form the feed and/ or precursor to innovation (Amabile, 1987; Dobni, 2008; Dobni, 2010; Zhuang, 1995). With creativity playing a key role at the individual level, at one end of the value chain, creativity by itself has multiple sources. Hence, if creativity can be seen as a dependent variable that is dependent on multiple independent variables such as motivation, skills, knowledge, perceptions, and attitudes (Amabile, 1987; Dobni, 2008; Zhuang, 1995; Caraballo & McLaughlin, 2012). According to Amabile (1987), creativity does not carry an absolute value, instead creativity is a continuum, which is influenced and driven by individual

motivations, skills, knowledge, and experience. Thus, increasing an individual's knowledge and experience, either by tacit or explicit knowledge, training or experience, has a positive impact on creativity (Dobni, 2010). According to Zhuang (1995), creativity is a dependent variable that is influenced by various independent variables and factors such as: a stimulating environment, high risk-tolerance, knowledge, skills, attitude, age, gender and job function. Sorting out the commonalities of knowledge and skills, identified in the research by both Zhuang (1995) and Amabile (1987), both author's research differs on the independent variables motivations versus attitude and high-risk tolerance at the level of the individual.

As Amabile (1987) noted, expertise is one of the precursors or influencers to creativity and hence, domain related expertise that is specific to a certain domain and gained via knowledge from trainings or experience could influence expertise and thereby influence creativity. Knowledge could be tacit or explicit and hence expertise could be composed of both tacit and explicit knowledge. Explicit knowledge is codified in nature and hence can be easily transferred, however tacit knowledge is gained by the individual as result of action, experience, etc. and hence becomes unique to the individual. The primary feeders for tacit knowledge are experience and application of explicit knowledge to situations that are rather challenging or unique as they come by. As a result, codifying tacit knowledge is a challenging and difficult task. In applying heuristic approaches to problem solving, with the direct proportionality between increase in diversity and increase in knowledge (Shin et al., 2012), adding diversity could lead to increasing creativity.

While creating workgroups or teams as part of the iterative creative process increases diversity. The iterative nature of the creative process make it necessary to be reviewed and monitored periodically by an individual or a group or a team of individuals (Amabile, 1999;

Zhuang, 1995). Diversity in workgroups brings variety to the knowledge pool and quantity of the knowledge at hand, thereby benefiting the iterative creative process by providing multiple perspectives from various standpoints/ perceptions that may be unique from members of the teams and vary by diversity in terms of culture, gender, experience, etc. An increase in diversity could lead to interpersonal conflict, and such conflict could negatively impact creativity.

However, such conflict could be mitigated with a clear understanding of the goals, purpose and vision and the conflict may be decreased by increasing the cognitive diversity, which increases task-conflict and thereby reduces negative effects of diversity and increases creativity (Shin et al., 2012). In rowing a boat, all members of the rowing team have to row in the same direction with a clear understanding of the direction and goals. Similar, in high performance teams, clear understanding of the team members in terms of the direction, purpose and goals is crucial. Such understanding is gained by clear focus and crystal clear communications (Sadri & Condia, 2012). A free-flowing and unstructured business or organizational structure, in some cases breeds and most definitely doesn't help in alleviating the negative impacts of diversity and hence shunts creativity. Thus, in attempting to tactically manage and mitigate the negative effects of increased diversity, formal structures are needed, along with leadership that carries intestinal fortitude and transformational attitude, with strategic foresight to create and sustain positive and stimulating work environments that feed on and value cognitive diversity and collectively help attenuate the negative effects of demographic diversity (Sadri & Condia, 2012; Shin et al., 2012).

The aforementioned research saw a role for the organization in influencing the drive for innovation, in addition to the individual's role on creativity. However, the authors of these studies differed on the importance and specific responsibilities of the organization as it applied to respective roles towards innovation. According to Amabile (1987), the organization played a

crucial role in collaging, organizing and making knowledge in a codified, accessible and user-friendly manner. The organizations role also included, ensuring the maintenance of a positive work environment that increased and positively influenced creativity for individuals in a team and also predicting the types of innovations the organizations would deliver as an outcome Amabile (1987). On the other hand, Zhuang's (1995) research pointed towards the organizations role to be critical and focused on creating policies and procedures that would enhance and help improve products/ services, productivity, and sales, by contributing towards a stimulating work environment that has high-risk tolerance, while ensuring that such policies and procedures align with a positively impact and motivate individuals and their behaviors. In addition, Zhuang (1995) and, Caraballo and McLaughlin (2012) saw creative output to be dependent on how individuals perceived innovation.

Carrying, transforming and realizing ideas to deeds requires the organizations to have the ability to manage creativity and the outcomes as of successful creativity outputs, which ultimately influence and grow innovation. Every organization has to manage tensions, and the key to manage the right tensions, while balancing profitability and growth, short-term and long-term goals, and looking for synergy across the organizations or focusing on specific teams alone (Dodd & Favaro, 2006). In assessment, such factors suggest the organizations culture on openness and flexibility to deal with progressive, yet dynamic and ambiguous environments. For creativity o flourish, sustain and grow, in addition to various organizational and individual factors, a supportive and integrative environment and infrastructure are needed. Although, the organization needs to play a key role in attracting, retaining and managing the organizations human capital; support, thought and pragmatic action are needed to ensure and incentivize desired creative and collective innovation behaviors that are supported via an infrastructure of

policies, procedures and organizational structures, which while providing support, also capture and incorporate newly discovered knowledge (Amabile, 1987; Zhuang, 1995). In the efforts towards acquiring and updating existing knowledge, organizations can follow multiple strategies, such as hiring new talent, mergers and acquisitions, collaborative alliances and partnerships with multiple sources for knowledge sharing, and investing in research and development facilities (Díaz-Díaz & De Saá-Pérez, 2012; Jusko, 2008; Sebestova & Rylkova, 2011; Wilson & Doz, 2011). The tactical approach for strategic knowledge acquisition depends on the organizations. Overall strategic goals, culture, the specific type of knowledge being sought, and most importantly, the organizations appetite to assimilate such knowledge.

Knowledge Management

Data are a collection of discrete facts that when put in context and analyzed translate to information (Hoe, 2006). A higher level of comprehension of information is knowledge (Davenport & Volpel, 2001). In determining the dichotomy between knowledge and information, Nonaka and Takeuchi (1995) noted that knowledge is about beliefs, commitment and perception, and is valuable towards achieving an end. Information on the other hand is a simpler form of knowledge. Since, knowledge is about perceptions and beliefs, there is variability in comprehension of knowledge at the individual level. According to Bierly, Kessler, and Christensen (2000), individuals vary in their ability to effectively choose and apply knowledge in a given context or situation. Individuals are unique in their respective abilities and approaches to solve problems (Haapasalo & Kess, 2001). How a person defines a problem drives the approach to solve the problems and hence the direct correlation with creativity and relationship to innovation. Individuals derive solutions to problems using their experience and knowledge, which might be tacit or explicitly acquired. Nonaka and Takeuchi (1995) presented

the SECI model on knowledge management that identifies four core processes, namely socialization, externalization, combination and internalization. The greater the experience, wider the toolset of knowledge and hence greater the alternatives solutions. Hence, an individual with greater experience could potentially be more creative as compared to someone with lesser experience and exposure with problem solving. According to Caraballo and McLaughlin (2012), perceptions of innovation vary in individuals. Depending on perceptions of innovation, individual's creativity may be impacted from experiences in an accept or reject form of decision, which in turn may act as a differentiator or a filter that either enhances or impedes creativity, and thereby impact the outcome of innovation.

Knowledge Management's Need

By combining and leveraging the synergies that emerge out of the combination of data, information processing and individual creativity, organizational processes formulate and feed into knowledge management and manage risk in an efficient manner in dynamic and ever changing environments (Haapasalo & Kess, 2001). According to Haapasalo and Kess, (2001), an interactive translation and capturing undergoes between explicit and tacit knowledge as business process help create, disseminate, renew and apply knowledge (Haapasalo & Kess, 2001). Although unique to the individual, tacit knowledge, by its interaction with explicit knowledge in an interactive manner could be captured and translated to explicit knowledge over a period (Haapasalo & Kess, 2001). Due to increased uncertainty in the value chain, and in order to manage uncertainty in a flexible manner, organizations shifted from an information-based to a knowledge-based operational style and basis for respective value chains (Haapasalo & Kess, 2001). An action that can be repeated to solve a problem that is recursive, could use explicit knowledge but such an environment is dependent on certainty factor. In an increasing dynamic

environmental and ecosystem, certainty shifts to uncertainty and hence, instead of approaching problems an actions with best practices, the focus is shifted to enhancing creativity. Despite the study conducted by Haapasalo and Kess (2001), the authors did not connect their conclusion to any specific type of innovation.

According to Senge (1990), there are five characteristics that a learning organization exhibits: systems thinking, personal mastery, mental models, shared vision, and team learning. Senge (1990) noted that learning organizations needed to foster a connected environment to facilitate interconnected mediums of thinking. Greater commitment from individuals, as employees, results in stronger communities in the organizations and allowing for accelerated organizational learning. Systemic interrelationships run deep in organizations and the acceptability and level of organizational learning has the roots set at the individual level, thereby making the case to ensure a shared vision and inclusive team or community engagement occurs for enabling organizational learning. The organization needs to align knowledge management with the overall strategy to achieve strategic goals in achievement of the vision and execution of the mission of the organization. Socialization and internalization processes of the Nonaka and Takeuchi's SECI model directly connect with informal knowledge processes of an organization (Hoe, 2006). Because of the codified nature of explicit knowledge, a knowledge management system within an organization, make it possible for the organization to capture, impart and update explicit knowledge. As a result, organizational learning focuses on recording, acquiring, assimilating and integrating tacit knowledge into explicit knowledge and ultimately into the organizations existing knowledgebase (Haapasalo & Kess, 2001; Mohanty & Kar, 2012). Overall organizational learning is dependent on various techniques and processes for information acquisition, such as experimentation, observation, research and development (Mohanty & Kar,

2012; Sirmon, Hitt, & Ireland, 2007). Factors such as the type of knowledge sought, management of the tensions between profitability and growth, short-term versus long-term, organizational versus silo (Dodd & Favaro, 2006), the organization appetite to handle, accept and assimilate new knowledge, and the organizational overall strategy, all collectively impact the specific techniques and tactics to acquire and assimilate new knowledge. Transforming information to knowledge, in order to increase flexibility and improve the organizations competitive stance in the market, also feed into the process of acquisition and dissemination of knowledge (Mohanty & Kar, 2012; Sirmon et al., 2007). According to Argyris (1998), organization could sustain or improve their operating status as it stands by leveraging organizational learning. Argyris (1998) noted that organizational could use single loop learning to introduce and assimilate innovations that are new or improvements, while the double loop learning would be most suited for change innovations that would translate to diffusion or adoption of innovations (Argyris, 1998). Single loop learning is used to maintain, while double loop is used to improve current operations, products or services. Individuals, via their job function, which is driven by or enhanced by their knowledge and experience, drive creativity, which in a collective and interactive environment enhances innovation, and the perceptions of innovation, as defined psychologically by the individual and influenced by the policies and procedures of the organizations, collectively direct to the innovation outcome that furthers the organization's strategy to sustain operations and remain competitive in the market.

How individuals define innovation seems to affect the learning processes and structures that collectively support and enhance a firm's overall strategy and competitive position (Argyris, 1998, Zhuang et al., 1999, Caraballo & McLaughlin, 2012).

Types of Knowledge

According to Wilson and Doz (2011), explicit knowledge, embedded knowledge and existential knowledge were the three types of knowledge that were identified to be maintained and the acquisition, processing and transmission depended on the type of knowledge. Wilson and Doz (2011) defined, explicit knowledge to be knowledge that can be codified, defined and transferred via common languages and processes (Wilson & Doz, 2011). Knowledge that is context-related, observable, loosely defined and can only be accessed when seen through different eyes was embedded knowledge (Wilson & Doz, 2011). Finally, knowledge that is context dependent, systemic, exists in behavior and norms, and the only way to learn is by doing was categorized as existential knowledge context dependent, systemic, exists in behavior and norms, and can only be learned by doing (Wilson & Doz, 2011). Embedded and existential knowledge collectively formulate tacit knowledge, where in embedded knowledge is partially tacit, while existential knowledge seems to be tacit (Wilson & Doz, 2011). Due to the codified nature of explicit knowledge, such knowledge can be transmitted and translated using existing, common language systems and thus has a common basis of understanding. On the other hand, tacit knowledge is context and individual driven, and because contexts and individuals are unique from a comprehension and perception standpoint, tacit knowledge requires greater effort for understanding and hence is challenging to be captured and synthesized using existing common knowledge system (Wilson & Doz, 2011). Attracting, foraying and experiencing are three techniques to access knowledge and each is applied to a varying degree depending on the type of knowledge desired (Wilson & Doz, 2011).

Transmission speed and quality of new information are impacted by a number of factors (Díaz-Díaz & De Saá-Pérez, 2012). Most factors would be categorized into two section: perceptions and culture. From the perception standpoint, uncertainty that involves validity and

reliability, and lack of perceived need for new knowledge or to ensure the maintenance of status-quo, are one set of factors that impact the transmission and quality of new information. Another major set of factors are culture driven and dependent on the willingness to accept externally sourced knowledge (Díaz-Díaz & De Saá-Pérez, 2012; Birtchnell, 2011). For instance, in rural India, due to cultural disbelief and a tendency to maintain status-quo, external multi-national markets, find it challenging to penetrate the market, even though there is a gap that could be filled. Mergers, acquisition, and or joint ventures, afford organizations the opportunity to purchase new knowledge, which may be perceived as positive by the organization's management, as the knowledge asset become an internal entity. Although perceived as positive by the organization's management, internal and prior organizational culture might demonstrate resistance and cause challenges to the newly acquired information from external sources (Díaz-Díaz & De Saá-Pérez, 2012). This becomes especially true in the case of tacit knowledge and perceptions being diverse and varied, without a common group open the knowledge up for discussion. In addition, the complexity or the need to integrate the newly acquired knowledge into an existing system or knowledgebase might need time and further effort (Díaz-Díaz & De Saá-Pérez, 2012).

In addition to being crucial for innovation, idea generation is particularly enhanced as a result of knowledge sharing (Naranjo-Valencia, Jimenez-Jimenez, & Sanz-Valle, 2011). During the idea generation phase, when exploring new areas and contexts, there is particularly an enhanced sense of uncertainty. Exploration does not flourish while staying within structures, and hence formal rules and structures tend to inhibit ideation, whereas fully free and unstructured sessions lead to ideas that may not be of value to the organization within the current strategy and vision.

The most effective access mode for accessing knowledge depends on the type of knowledge, which along with existing knowledgebase, organizational culture and structures of the acquiring firm will drive absorption of the knowledge (Díaz-Díaz & De Saá-Pérez, 2012). Given the uncertainty and dynamism in the value chain, knowledge acquisition strategies seem to have shifted from historical content-based system to one of flexible content-based systems (Haapasalo & Kess, 2001). Although existing resources in knowledge management systems of an organization might provide sufficient information for the organization to operate with the knowledge needed for new or improved innovation, the increased level of uncertainty and constantly changing technology along with rapidly transforming innovation, necessitate the organization to look for acquiring knowledge from the outside (Díaz-Díaz & De Saá-Pérez, 2012). Given the focus on tacit knowledge to deal with increased uncertainties, changes to the type of knowledge seem to receive greater importance weighted on the human resource capital as opposed to technology, although technology is the platform to carry, synthesize and port such knowledge, while the source of such knowledge is maintained from human resources. In doing so, the wider definition of the type of knowledge that covers perceptions and culture, provides for definitions that closely align with specific definitions of innovation.

Acquisition of Knowledge

Globalization provided a global platform and in a sense a flattening of the earth, as result there is an increase in dispersion and diversity of sources for various knowledge, which does increase the challenges from a perception, culture and acceptance standpoint, while providing a much broader and deeper variety of solutions for innovation outcomes. The precursors to idea implementation, with the existence and sequencing of the idea generation, themselves provide

for at least two different environments with different characteristics that drive innovation (Bouncken, 2009).

According to Sebestova and Rylkova (2011), a strong link exists between innovation and creativity and non-traditional thinking. If non-traditional thinking is attributed to out-of-the-box thinking, then this would fall under the category unique attitude that an individual would bring to enhance creativity. The easier knowledge is to transmit, absorb and assimilate, the quicker is the adoption by individuals in an organization. Building on this platform, Sebestova and Rylkova (2011) noted that knowledge enhancement at an organizational level starts with focusing and ensuring, increased facilitation of ease of knowledge transmission and assimilation.

Acknowledgment and acceptance are different from one another. Sebestova and Rylkova (2011), noted that willingness to change was more important than the learning process, although organizational learning was key, because lack of willingness will not provide the organizational momentum to adopt and change using the newly acquired knowledge. Hence, a formal process may be justified for knowledge acquisition and management in order to facilitate the use of knowledge and adoption of the knowledge by the organization, as required by the need for organizational learning and willingness to change.

According to Sebestova and Rylkova (2011), the occurrence of learning takes place at all levels of the organization structure within the organization, maintaining a reinforcing relationship between learning and knowledge management. Although, organizational learning is identified as a need to sustain and increase the organization ability to innovation, and ensure competitive advantage for the organization, there lacks a consistent correlation between learning and successful innovation. This lack of a consistent correlation further reinforces the need for a strong relationship between the organization's business model and strategy. Thereby suggesting

that organizations with effective learning techniques and adoption need effective management processes, policies and structures to transform innovations in services and products to marketable services and products (Erwin, 2002).

Boundary-Spanning Activities

The sources of information, internal or external, and the information needed, depend on the how the organization and/ or the individual define innovation, new, improved, or change, and what their perception is of innovation. The definition and perception of innovation for organizations that consider themselves to be leaders or disruptors is different from the perception of innovation by organizations that think of themselves as the fast second or follower or imitators.

Alignment with external organizations or across organizational boundaries with t=in the organization involving various activities of representation, coordination, sharing, and searching that focusses towards different audiences and purposes encompasses boundary spanning (Marrone, 2010).

In environments that are stable and seeking collaboration with groups that are more powerful or knowledgeable, representation is found to be useful. Thus, making representation a viable technique for spanning explicit knowledge (Björk, 2012); Marrone, 2010; Gurbani, Garvert, & Herbsleb, 2010). While seeking tacit knowledge, searching is beneficial and hence, searching is a viable technique to be used in environments that are unstable, dynamic and volatile (Cummings & Pletcher, 2011; Tuulenmaki & Valikangas, 2011). Coordination activities finds it position to be a balance between representation and searching (Marrone, 2010). Critical decisions in boundary spanning involve decisions around resource allocation, team formulation, leadership on teams, and network infrastructure and platforms for seamless information t6transfer

and exchange (Marrone, 2010). The efficiency of boundary spanning as applied to the enhancement of information transfer and improvement of innovation depends and falls on the team and the individuals who make up the team, their respective behaviors and the effectiveness of the network (Marrone, 2010). Ultimately, the need for efficiencies is driven by the understanding and perceptions of innovation the individuals and teams, which influences behaviors at the individual level (Marrone, 2010; Caraballo & McLaughlin, 2012).

A correlation exists between the rate of innovation and the degree of external collaboration (Kang & Kang, 2009; Sawang & Matthews, 2010). Prior research shows that open alliances, informal networks, research and development (R&D) collaboration, and technology acquisition, provided stimulus for activities for external innovation (Boudreau & Lakhani, 2009; Chesbrough & Garman, 2009; Kang & Kang, 2009). In addition to innovation, other factors such as focusing on the core business, transitioning R&D efforts to move external, lowering operational costs, etc., were also dominant drives for such alliances and actions. From the strategic perspective, managing the right tensions between profitability and growth, short-term versus long-term goals, organizational versus silo or team focus were the primary drivers and motivations. (Dodd & Favaro, 2006).

Innovation does not follow a pattern and can be of complex and non-routine in nature, encompassing individuals, teams, product leads, pilots, experiments and external sources from other organizations, customers, stakeholders, etc. and hence the need for flexible approaches when dealing with risk and uncertainty (Cummings & Pletcher, 2011; Gurbani et al., 2010; Tuulenmaki & Valikangas, 2011). Although the researchers agreed on the commonality that such characteristics and techniques were leveraged in development of new products or services, new business models, and off-the shelf products, they all disagreed and differed on which of those

characteristics were most important. What is unclear is if such variation is due to or driven by the type of innovation or a result of the organization and its culture. According to Gurbani, et al. (2010), management's role in establishing and clearly articulating and communicating the idea implementation criteria is a significant innovation driver.

Alliances

Innovation alliances are of two types: open innovations and clusters of innovation. According to Sebestova and Rylkova (2011), a correlation and synergy effect exists between clusters of businesses that share knowledge and qualitative variables unique to each organization that drive as to how well knowledge is assimilated and applied. Knowledge networks and ecosystems came about as a result of alliances that were formed by informal relationships (Sebestova & Rylkova, 2011). As part of the research, Sebestova and Rylkova (2011) developed a five-step competency model that could be used for sharing knowledge and provided a framework for successful alliances: care, competitiveness, communications, culture, and clarity on the relationship. This competency model maintains that alliances are formed over a period and, develop and progress depending on strengths, weaknesses and needs of one another. As the variety and diversity of the ideas widens, so does the diversity and size of the alliances, thus suggesting a direct relationship between exposure to ideas, and size and diversity. Since the alliances develop over a period, the alliances also develop a mutual understanding of trust and comprehension of context, thus resulting in smoother transference of tacit knowledge as the alliance or relationship matures (Brettel, Heinemann, Engelen, & Neubauer, 2011).

Large organizations and high technology forms prefer open alliances, largely because most of the common base of knowledge already exists in the form of standards and hence, protecting core competencies or core technologies is not an issues, whereas, small organizations

and low technology organizations prefer mergers and acquisitions, just to ensure the core competency is protected and governance is not an issue (Hagedoorn & Duysters, 2002; Han, Oh, Im, Oh, Pinsonneault, & Chang, 2012). Hence the two key factors that influence alliance decisions are governance and protection of existing capabilities or technologies. Expansion of open alliances results in issues with governance and make it challenging to protect the core capabilities, thus negatively impacting the organizations innovation alliances.

Open Innovations

An innovation alliance that involves multiple external partners and is used to decrease the development time or cost of the new product (Brettel et al., 2011), and share knowledge and improve innovation by accessing and transferring information from a diverse and wide array of sources is called open innovation (Jusko, 2008; Lasagni, 2012). An example of such collaborate open innovation is Tata Nano (The Economist, 2009), the cheapest car in the world (Ray & Ray, 2011). In developing the Nano, an innovative modular design was developed following the open innovation alliance between the Tata Motors groups and suppliers of parts and service (Ray & Ray, 2011). Through a diverse array of modalities, open innovation might also encompass informal alliances, clusters of innovation, and customer involvement, although these are not venues that most firms would pursue or are ready to be involved. At the benefit of not having to reinvent the wheel, affiliations with external sources can afford a wider array of knowledge at a lower cost. However, the trade-off comes in the form of an increased possibility of losing intellectual rights and challenges with accepting and adopting tacit knowledge (Lasagni, 2012). Policies, procedures and structures are required as part of boundary spanning as the individual sources at some point will need to transfer information to other participants or team members, internal or external, in order for the idea to transform to a deed or to enhance the innovation via

alliance (Hoegland, Lichtenthaler, & Muethel, 2011). This is the transfer point when individual perceptions come into effect and could dramatically progress execution of the strategy or turn into an obstacle or roadblock hindering further progress on the innovation.

According to Rufat-Latre, Muller and Jones (2010), there are two components to open innovation as applied at the organizational level: gap analysis to identify the delta between the current capabilities of the organization versus potential growth areas and shifting focus to enhancing the core capabilities as opposed to focusing on the market-share. Gap analysis affords the organization to become effective and efficient and strengthen the core, even though such analysis may not lead to any strategic changes or value. However, such an analysis and implementation can expose hidden or un-tapped talent, knowledge or capabilities that the organization can deliver on. Such analysis helps with changing existing beliefs, mindsets and internal structures or hierarchy (Henttonen, Pussinen, & Koivumäki, 2012). Changing culture is the most challenging and the accepted norm of bringing change is by starting with changing policies, followed by procedures then beliefs and then culture (Christensen & Overdorf, 2000). As Christensen and Overdorf (2000) noted changing the culture is the most challenging and time taking task that is also disruptive in nature. In such scenarios, gap analysis can help with setting in incremental changes that would start with scanning and identifying the gaps and incrementally making changes to shift the culture. Gap analysis helps identify potential growth areas where as shifting the focus from market share to core enhancements helps increase the possibilities for delivering change innovations. Alignment of the organization's structure, processes, beliefs and culture with the company strategy is a requirements in both cases. Rufat-Latre et al. (2010) noted that innovation is a process and though open innovation evolves and matures over time, the extent of the organization's participation in an alliance and the level of innovations depend on

the organization's culture and comfort level with the uncertainty and risk, willingness of the organization to change its process, structures, strategies, which in turn drives the benefits that can be derived (Srivastava & Banaji, 2011).

According to Wilson and Doz (2011), alliances had the potential for greater success and acceptance in instances where the companies were equals in areas that were separate from the purpose or goals of the alliance. Open innovation is part of a holistic and broader perspective that may be formed by dynamic business models, adopted or dissolved in reaction to market demands, need for knowledge creation and the core need for innovation (Wilson & Doz, 2011). Alliances may be viewed as dynamic in the sense that both partners need to bring to the relationship some form of an asset that is unique to their respective organizations and seek each other out form a symbiotic relationship that builds on the target organizations platform, ultimately both organizations emerging to be stronger and more competitive.

Customer Involvement

In search of knowledge and solutions, boundary spanning could also involve customers, which can participate via feedback and pilots towards devising solutions. Involving customers in the design process allows for innovations that help with reducing costs, meeting the exact needs of the customers, reduction in cycle time, and reduced risk; although, the same exposure could lead to instability and high risk due to exposure and changing technology, while at the same time setting a perception that the organizations is catering to the needs of current customers with new and improve innovations, but not to focusing on the un-tapped potential of new customers and new markets that might need radical or change innovations (Bower & Christensen, 1995). First movers with market orientation receive an upper hand and a competitive advantage over fast-second or slow followers, however, such an advantage is short-lived due to the repeatable nature

of explicit knowledge that delivers new and improved innovations (Kumar, Jones, Venkatesan, & Leone, 2011; Sandmeier, Morrison, & Gassmann, 2010). Identifying lead users helps refine the product but at the same time sheds and limits a laser focus on development of the product or service that may not be inclusive of all the needs of a wide customer base, thereby making the product or service to be geared towards a niche market (Sandmeier, Morrison, & Gassmann, 2010). Market orientation cannot be short term and must be maintained as a long-term goal, much more in terms of cost of doing and staying in business as opposed to the need for competitiveness (Kumar, Jones, Venkatesan, & Leone, 2011).

Two information gathering methods, on-line competitions (open approach) and focus groups (closed approach), are used to involve customers as part of the boundary spanning process and to gather knowledge for the innovation process. Each methodology has its own pros and cons, along with general applicability and customized use for specific phases of the innovation process (Schweitzer, Buchinger, Gassmann, & Obrist, 2012). Because new and improved innovations directly relate to existing products or services, a closed approach is recommended (Schweitzer, Buchinger, Gassmann, & Obrist, 2012). As for radical innovation a combination of closed and open approach is recommended, with the idea generation phase going through a closed approach and open approach for the later phases, given the level of uncertainty and risk involved in radical innovations. In addition, online competitions as applied to radical innovations help ensure a lower cost, wider diversity and higher levels of creativity.

The two most noted reasons for the degree of participation in an alliance are protection of intellectual property and governance. The risk and associated concerns apply to all organizations, while external boundary scanning can involve multiple partners and take various forms (Abuhamad & Shaltoni, 2013). Even though risks exist, new knowledge is a requirement for

innovation and without innovation, the organizational cannot sustain in a market or continue to stay in business. Thus, the need for every organization to balance the risks (Brown, 2010) and attempt to acquire and disperse new knowledge. While there is a risk due to the voluntary nature of alliances and motivations of participants, there is also a risk from the knowledge that is transferred or acquired as part of mergers or acquisitions. The uncertainty of the organizational culture and its influence on the will to act on the newly found knowledge adds to the uncertainty and predicament of gaining new knowledge. The state of affairs, resource availability and uncertainty of the environment may limit the available options, final decision must be dependent on ensuring the alignment of culture, processes, company strategy and infrastructure with the perception of innovation at the organizational and individual level (Rothaermel, & Hess 2010).

Clusters of Innovation

Engel and del-Palacio (2011) identified, as part of a theoretical model of clusters of innovation (COI), emphasis on international perspective, entrepreneurial activities, and symbiotic relationships as key to the success of an organization. Depending on the strength and trust of the relationship and interdependence, behavior and commitments of participants varied along with complexity of the relationships, which translated associations over a range wherein the clusters of innovation varied from network clusters to super clusters. COIs require a level of trust and startups can be the greatest beneficiaries of COIs as a result of the network and access to knowledge, although organizations of different and varying sizes and maturity levels might enter into partnership (Engel & del-Palacio, 2011). Even though the COI theory seems to embody the organic nature of relationships and shows uncertainty from innovation. There is lack of clarity in terms of how the model operates and how and to what extent knowledge sharing takes place while protecting intellectual knowledge. In COIs, while respecting competitive

advantage and conflict of interest, given the risk to intellectual property, mutual interdependence is pervasive with explicit knowledge with low and restricted access to tacit knowledge (Audi, 2011). The likelihood of forming innovation alliances is greater during times of uncertainty and idea exploration as opposed to stable and idea exploitation phases (Audi, 2011). According to Audi (2011), financial resources and strategic objectives were some of the dominant factors in forming geographic clusters, in addition to knowledge transfer as a primary motivation. The degree of variation in alliances suggests that searches for explicit knowledge become the dominant need as part of the collaboration facilitation, although welcome, tacit knowledge may be viewed in a less favorable sense due to the increased levels of uncertainty and turbulence. More than geographic proximity, technological proximity influenced to a greater extent the formation of clusters of innovation (Audi, 2011). Given the redundancies of information sharing within a locale, Audi (2011) identified the existence of a ceiling point or upper limit on the number of organizations that formed COI, thereby suggesting a tipping point beyond which the COI sees a decrease in density. The study by Audi (2011) also identified that the tipping point at which local memberships decline, there is an increase in international memberships, hence suggesting that the tipping point is the point of departure for increase in international alliances, which also points to the possibility of knowledge saturation, lack of diversity or need for greater sophistication in the locale or increased growth of the organization causing the search of new knowledge from other diverse sources beyond the local region (Audi, 2011). Alternatively the search may be driven as a result of dissatisfaction from an existing COI.

Culture

Culture is formed as a result of combination of perceptions, beliefs, interactions and societies that people inherit, develop and sustain over time. Assumptions, trust, beliefs and values are created amongst the individuals in a culture as a result of coming together to solve problems and overcome challenges. Functioning at the national, organization and individual level, culture, becomes visible thru the actions, institutions and structures that individuals build, accept and adopt. In creating a value system on shared assumptions, culture helps identify behaviors and beliefs that are desirable, supportive of the surroundings and environment that individuals are in and attracts new members into the society via the socialization vehicle.

Culture at the National Level

Culture can either consume strategy and be the impregnable roadblock or be the unifying force that empowers decision making by situating the individuals in context to act in the habitual traditional way or the innovative, out of the box manner to solve and deal with challenging situations (Kaasa & Vadi, 2008; Scroggins, 2006). Individuals work within the policies, procedures and structures that organizations and society put in place and hence the environment needs to be stimulating to foster creativity and to propel innovation. Culture can help tune and condition attitudes and behaviors via the affiliation and leverage of risk and rewards, thereby directly impact innovation (Kaasa & Vadi, 2008; Williams & McGuire, 2010).

Societies are made up of subgroups and culture is a characteristic that differentiates one sub group from the other within the larger construct of the society (Hofstede, 1980a). Although the subgroups via their unique characteristics could be differentiated from one another, the society, that includes all the subgroups has characteristics that permeate and are common through all the sub groups within the society (Hofstede, 1980a; Williams & McGuire, 2010). Culture is

formed as a result of formal and informal learning by individuals and sub-groups with common beliefs, a system of trust, common traits, and commonalities in terms of approaching problem solving and conducting business (Hofstede, 1980; Kaasa & Vadi, 2008). . Culture affords predictability of behavior, however, culture by itself, is not tangible and could be measured as a tangible only via the behaviors, institutions, structures and systems that the individuals and subgroups in that culture exhibit (Hofstede, 1980a; Williams & McGuire, 2010).

The process of innovation involves creating and devising something that is new, improved or a change, in order to solve an existing or new problem or to fill a gap that has been identified. And if culture can help characterize or predict behaviors of individuals and subgroups in how they approach problem solving, then there is a direct relationship between culture and perception of innovation, with culture directly influencing innovation. The initial research conducted by Hofstede (1983) identified power distance, uncertainty avoidance, masculinity-femininity (gender), and individualism-collectivism, as the four cultural dimension at the national level, which later saw long-term orientation added to the list (Hofstede & Bond, 1988).

The economic growth in the Asian countries (tiger economies), followed the identified characteristics of high power, high uncertainty avoidance, masculinity, high collectivism and long term orientation, on the other hand, the opposite trends were favored innovation in the western countries (Franke, Hofstede, & Bond, 1991; Hofstede & Bond, 1988; Kaasa & Vadi, 2008). Such differentiation points out to the existence of additional factors that may be at play.

Organizational Culture

An integrated and intertwined relationship exists between the national and organizational cultures, both of which carry one another traits to varying degrees with factors of commonalities, common beliefs, types of ownership industry, competitiveness, purpose and diversity (Leung,

Bhagat, Buchan, Erez, & Gibson, 2005; Yim-Yu & Nicholson, 2008). Increasing complexity of organizations and technological advancement leads to increase in the organization's susceptibility of accidents, thereby impacting reliability (Weick, 1987). Similar to national cultures, organizational cultures also carry similarities in terms of the function that ensure stability, continuous operation, competitiveness and identification of acceptable behaviors, beliefs and thought processes (James, 2005). Weick (1987) suggested examples of communication deficiencies in pilots training on new planes and communication systems at NASA, where the individual, while learning via simulation or training adapts to new processes, but errors or susceptibility to accidents arises not just in the manner in which such knowledge is acquired but also from perceptions and habitual behaviors that see the new process or innovation as a change from an individual perspective. As in the innovation continuum, consistent behaviors vary and drift between opposite poles of tradition and innovation, while encouragement and empowerment of creativity becomes part of the organizational culture (Kenny & Reedy, 2006). Thereby, the innovation strategy of the organization is greatly influenced by the organizational culture (Naranjo-Valencia et al., 2011).

Stability or instability are characteristics that are experienced by all cultures and as such are influenced by severity and duration as the dominant characteristics, which in turn can help influence and predict the approach to be traditional or innovative when solving problems and also identify which of the approaches is most successful. The exact similarity may also be found in economic environments as it applies to stability. For instance, economic environments that are stable tend to be low risk-taking, are usually reactive, emphasize on efficiency, identifying the gaps and getting more and more efficient, without disturbing stability. On the other hand, unstable environments are high risk-taking, proactive, willing to try radical changes with high

risk tolerance (Leung et al., 2005). Hence, new and improved innovations seem to flourish in stable environments, whereas change and radical/ disruptive innovation seem to flourish in unstable and high-risk tolerant environments. In this context, it is important to note that every product and service has a shelf-life and the process of evolution, maturity and market focus never stay constant and are constantly changing. Hence for an organization to be successful, always operating in a stable or only operating in an unstable environment may not be conducive and a balanced approach of coalescing stable and unstable environment would be a more pragmatic and beneficial strategy to a sustained competitive advantage.

The Role of Organizational Culture

Resource utilization could be maximized as a result of operating in a dynamic environment, with agile deployment of products and services, instant dispersion of knowledge and information, and flexible knowledge acquisition and execution strategies (Wilson & Doz, 2011). The key in these characteristics from the culture is the organizational ability and willingness to execute, accept and assimilate new knowledge, which by far, transforms an idea to a deed. According to Wilson and Doz (2011), structural and cultural changes within an organization can help achieve the combination of both the acquisition of new knowledge and new innovations along with executing and accepting on the new knowledge. The research conducted by both, Sebestova and Rylkova (2011) and Wilson and Doz (2011), identified the importance and criticality on the willingness to act, accept and execute, however, Wilson and Doz (2011), seemed to provide a holistic image on the approach to organizational learning and knowledge management as applied to innovation competencies and perceptions.

The policies, procedures and structures in place at an organization are to an extent driven by the organizational culture and, do require adjustment to execute on the organization's

strategy. Hence making cultural and strategic modifications, is necessary to shift and condition the internal processes of the organization and such actions can be influenced and managed by implementation of effective knowledge management and acquisition of knowledge from external alliances or open innovation. However, such structures and changes with diverse knowledge are only effective if there is the willingness to accept, change and execute on the new knowledge, while maintaining a clear focus on the market (Wilson & Doz, 2011; Sebestova & Rylkova, 2011). Hence, the “how” on achieve such change recommends the need for the organizational culture to be flexible and accepting to the extent possible.

Uncertainty and Its Impact

In determining the success of the innovation process, organizational culture plays a crucial and critical role, in addition to external factors such as market and customer demands, changes in customer trends and needs, which is extremely prevalent on the rural small and medium businesses (Kenny & Reedy, 2006; Lin, Chuang, Chang, & Yeh, 2012; Tellis et al., 2009). For instance, in catering to the needs of customers at the bottom of the pyramid in rural India, the key to provide products in inexpensive and small portions for consumption by a large group of customers. Following the trend, companies had to change their processes, packaging and marketing techniques to suit this need (Birtchnell, 2011; Ray & Ray, 2011). For radical innovation to take place, organizations need a stimulating, high-risk tolerance environment, while at the same time organizationally, the systems and structures should be clear about the uncertainty and high-risk factors. Absent such an environment and infrastructure, creativity cannot be sustained, which directly influences innovation. Often time, instead of seeing innovation as a holistic and all-organization practice, innovation is seen as the duty or job of a specific unit or department, R&D or just text slow and low-risk innovations in the mainstream

(Jamrog, Vickers, & Bear, 2006). It is valuable to separate the risks from day to day stable operations and test innovations in a sandbox, but the integrated infrastructure needs to be in place from an organizational standpoint to assimilate and absorb successful innovations. With diversity at the workplace and organizations operating globally, organizational culture plays a dominant role on the innovation process as compared to the national innovation with essentially becomes a degree of influence in the broader schema of factors that influence perception of innovation.

Change and Risk

The extent to which change is accepted or rejected is directly influenced by culture (Kenny & Reedy, 2006; Leung, et al., 2005). Change is not always welcome, in fact change is one such factor that can cause angst in individuals, disrupt the office harmony, increase uncertainty, cause instability issues, and shift power (Leung et al., 2005). Changing culture is the most difficult and needs to start with changing processes, behaviors, values, and beliefs (Hofstede, 1980; Christensen, & Overdorf, 2000; Leung et al., 2005). According to Hofstede (1980), behavior is influenced by the perception of what is acceptable. Existing thoughts and beliefs take an inordinate amount of time to change and hence, appropriate incentives could be leveraged to produce desired behaviors and changing values, which are reflected via the cultural influence and directly impact perception of innovation and acceptability.

Innovations and the process of innovation requires formal processes and structures that foster innovation Jucevicius (2010). According to Jucevicius (2010), a positive correlation exists between innovations, tolerance for risk, change, flexibility, learning at all levels of the organizational structure. And for improving trust driven boundary spanning. In resource constrained organizations, the resources needed for the innovation process are the same resources that maintain and sustain day to day operations. Hence, as a result of limited resources,

the burden is on the organization to judiciously use the limited resources available. In this context, while maintaining that innovation thrives in high-risk environments, Jucevicius (2010) notes that formal strictures and processes in the innovation process would actually ensure resources are used judiciously as opposed to the thought process that innovation need to happen in an unstructured manner. The process of innovation involves risk, uncertainty and explorations of those facets that have previously not been discovered and hence, having a formal structure and process ensures proper utilizations of resources while also capturing the lessons learned as part of the innovation process, which might fall under the category of tacit knowledge. In doing so, given the extent of uncertainty and risk, it may be advisable to maintain a separation in the resources used for sustain operations versus those dedicated for exploration. Change becomes much more manageable in unstable environments for these organizations that adopt the model suggested by Jucevicius (2010). National cultures take time to change as the decision making process depends on traditional approaches and, historic and habitual knowledge. Thus, countries cannot increase their risk in an unstable environment and be quick to adopt a change or afford a turbulent environment (Hofstede & Bond, 1988; Franke, Hofstede, & Bond, 1991; Jucevicius, 2010). If organizations can change rapidly is debatable and largely depends on the culture of the organization and level of creativity of the individuals, although the quickest change can be set-in at a time when the organization is facing a crisis, which provides the necessary stimuli and openness to adapt and adopt to new knowledge and practices.

National and organizational cultures impact and influence the individual while also influencing one another. An organization can be diverse with individuals from multiple nations and thus perceptions from a national culture standpoint, while at the same time national culture

can be made up of multiple organizational cultures. Thus, it is possible that creativity, innovation and economic success are influenced by factors, in addition to those noted above.

Discussion

Competitive advantage (CA) is different from sustained competitive advantage (SCA). Barney (1991) took a broad view of resources and brought about the differentiation between CA and SCA in the Resource Based View (RBV) model. CA provides with competitive advantage for the day to day operations, while SCA provides competitive advantage from a strategic perspective. According to Barney's (1991) view, SCA should be such that others/ competitors should not be able to copy or duplicate the resource and must be beneficial at the tactical level. Being unable to duplicate, provides the organization with long-term advantages. In assessing perspectives of innovation, the characteristic of being inimitable, identifies SCA as a criteria.

National Culture

Innovation is not unique to a specific national culture and innovation exists and flourishes in multiple cultures. In addition, innovation is dependent on condition of resource availability, geography, governmental policies and structures, etc. Being part of the same culture does not imply that an organization is successful. If innovation exists under such varying conditions, it is not possible to identify a single culture as the best or most appropriate culture for innovation. Even though national culture is a source for CA and SCA, the aforementioned arguments seem to ignore these factors.

Organizational Culture and Innovation

Culture is an amalgamation of interaction patterns of people, their beliefs, behaviors, practices assumptions and values. These are the factors that form the base and foundation of culture. With commonalities such as socialization and incentivization for desired behaviors, organizational culture and national culture share many of the common functions. The common beliefs, values and assumptions are visible in the policies, procedures and structures of organizational cultures. Hence, desirable behaviors and the creation of and organizational environment that incentivizes desirable behaviors are all identified by organizational culture. A critical component of creativity is the individual's task motivation, which is determined by the organizational culture that in essence established which type of creativity is permissible, incentivized, and accepted. Creativity on the other hand is crucial for innovation.

Socialization and collectivism aspects of the organizational culture work as a force of attraction and invitation for new members who share similar beliefs and values, exhibit similar behaviors, which in turn translates to shared assumptions on creativity and perception of I

innovation. For an organization to be successful, and because beliefs and assumption form the foundational baseline for organizational culture, the idea generation and idea exploitation aspects of the business need to be alignment, while knowing that idea generation happens at the individual level and idea exploitation is a collective process. Hence, we can conclude that organizational culture influences individual behaviors and their ability to be creative, which in turn influences the perception of innovation and impacts the strategy used for the innovation process.

Creativity can foster and flourish at the individual level provided the organization provides for the support infrastructure to support and permit creativity. According to McLean (2005), most research focused on creativity at the individual level, with little linkage to the broader organizational perspective. With creativity having the outcome of delivering something new and different is bound to introduce a change at the organizational level, there by suggesting a strong relationship between creativity at the individual level (idea generation) with innovation at the organizational level (idea exploitation). Hence the organizational culture could potentially hinder or be a catalyst for creativity or idea generation, and for innovation or idea exploration at the organizational level (Bouncken, 2009).

Socialization conditions and influences individual behaviors to be desired or undesired, while coordination enables those desired behaviors to execute at the organizational level with the policies , procedures and structures put in place by the organization. At the individual level, shared values, beliefs and assumptions show as the precursor and culture that influence the behavior of the individual, while policies, procedures an structures established by the organization apply towards the collective practices that are driven by culture at the organizational level (Donate & Guadamillas, 2010). The organizational culture, both, establishes

boundaries and incentivizes acceptable individual behaviors (McLean, 2005; Donate & Guadamillas, 2010). Thus, organizational culture can influence innovation with the coupled approach of socializing desired behaviors for idea generation and coordination of idea exploitation or acceptable activities as part of the organizational innovation process, leveraging the policies, procedures and structures established by the organization (Naranjo-Valencia et al., 2011; Tellis et al., 2009).

A multi-national study conducted by Tellis et al. (2009) on 700 firms identified a positive relationship between organizational culture and radical innovation. Of specific interest the researchers found that high-risk tolerance and shared values of flexibility were needed to influence and drive radical innovation, which was identified as a strategic goal. Naranjo-Valencia et al. (2011) conducted a similar research on 450 Spanish companies, using Cameron's and Quinn's (1999) Competing Values Framework (CVF), in an effort to identify the prevalent characteristics of the organizational culture's competitive strategies and study the relationship between innovation versus imitation. Organizations that executed on innovation strategies seemed to follow the adhocracy characteristic as opposed to hierarchy, which was dominant in organizations that followed imitation (Naranjo-Valencia et al., 2011). Adhocracy and hierarchy form opposite sides of the continuum, whereby adhocracy empowers flexibly and high-risk tolerance environments with an external focus and hierarchy, with the structural and protocol oriented characteristic is mostly suitable for low-risk stable environments. Hence, adhocracy is dominant in cultures that foster radical innovations and hierarchy is dominant in cultures that follow new and improved innovations. Exploration, uncertainty and high-risk stimulating environments drive radical innovation, while low-risk, incremental product and service extensions drive new and improved innovations, which in essence are tested and repeatable

practices and hence are leveraged by companies that follow an imitation strategy as opposed to companies that introduce disruptive innovation by leveraging radical changes. In either case, there is clear indication that organizational culture, which encompasses the shared beliefs, values assumptions along with the policies, procedures and structures, drives the strategy for desired type of innovation. Culture has its foundation built on common and shared beliefs, values and assumptions of individuals, who collectively form the organization. Hence, aligning the shared values of individuals towards risk, uncertainty, behaviors, policies, processes and structures within the respective organization is key to ensuring successful execution of the organizational holistic strategy.

Each organizational culture has varying degrees of various characteristics that manifest in terms of policies, procedures and structures that differentiate the organizational culture's approach to innovation. With such a variation at the collective level of the organization amongst organizational cultures, the variation has to be prevalent at the individual level also, because organizations are made up of individuals and the organizational culture is a manifestation of the shared values and beliefs of the individual in respective cultures. Hence, as varying organizational strategies illustrate varying definitions of innovation at the collective level, it is logical to assume that perspective of innovation must vary at the individual level as well. For an organization to be successful, the company needs to align the organizational culture along with the individual perspectives of innovation as part of the overall organizational strategy.

Individuals' perceptions influence the organizational culture and also define innovation depending on perception of innovation at the individual level. Hence, individual perception of innovation becomes the root for sustained competitive advantage for the entire organization.

Innovation must be new and useful and while there is a linkage between intrinsic motivations and creativity at the individual level, intrinsic innovation seem to lead to new ideas or idea generation but not necessarily beneficial or pragmatic in all cases (Grant & Berry, 2011). According to Grant and Berry (2011), intrinsic motivations could lead to novel ideas generated by curiosity, exploration and risk taking, however, the actions on those ideas are influence by the collective will and force of the overall organizational culture. Hence only those ideas that are influenced by the organizational culture move to idea exploitation and implementations phases. Hence, even though individuals may have creative and novel ideas, it does not mean every idea will transform into a deed and as such the process is, in essence, moderated and influenced by the organizational culture. Hence, an individual who aligns perception of innovation from an organizational standpoint and from their creativity standpoint, will find positive reinforce and feed back in moving ideas to deeds. Thus making the organizational culture the collective and cohesive force to identify, prioritize, socialize and coordinate the identification of novel ideas that would deliver useful innovation from an organizational standpoint, while originating at an individual level.

Research conducted by Donate and Guadamillas (2010) on Spanish industrial organizations showed that the process versus product innovation outcomes had a positive relationship with cultural emphasis on cost control as opposed to socialization. Type of knowledge and their acquisition strategies, acceptance, transmission and assimilation techniques are identified by organizational cultures. In addition, organizational cultures identify the type of knowledge and mechanisms of knowledge dispersion by aligning with other companies that were different in respective innovation strategies as applied to cost control versus differentiation in the product or service. Thereby resulting in further confirmation that innovation is viewed

differently by different organizational cultures, which align with the overall company strategy by establishing the requisite policies, procedures and internal structures.

Individual Creativity

Individual factors such as gender, job function, age, and attitude were identified as factors that influence perception of innovation and different types of innovation (new, improved or change) at the individual level (Caraballo & McLaughlin, 2012; Zhuang, 1995; Zhuang et al., 1999). In addition to the above mentioned factors the research conducted by Caraballo and McLaughlin (2012), Zhuang (1995) and Zhuang et al. (1999), also identified high risk tolerance at individual and organizational level and stimulating work environment with the necessary policies, procedures and structures, supported by the organizational culture as the forerunners for innovation. The literature suggests risk tolerance as a cultural characteristic at the collective level, however, the same risk-tolerance, in varying degrees, also persists at the individual level. Cultures that exhibit extended periods of instability or turbulence may have the capacity to endure higher levels of risk and thereby foster a culture of creativity that leads to change innovations as opposed to just new or improved innovations.

Boundary spanning techniques that target explicit knowledge are prevalent in individuals who are likely to operate in low risk environments, thereby leading to new and improved innovations. Whereas, individuals who are comfortable to work in high risk environments, are likely to be comfortable with boundary spanning techniques that target both explicit and tacit knowledge, and might be comfortable to lead into radical or change innovations. Individuals who are more adverse to risk may come from cultures that experience resource scarcity, which is a cultural characteristic that drives certain behaviors as part of the culture, at a national and organizational level. Every organization needs different types of innovation. New and improved

innovation that are incremental in nature help the organization maintain competitive advantage in the current market, while radical innovation provide companies with long term strategic advantages that ensure sustained competitive advantage. The focus of these different types of innovation suggests an organizational role, via policies, procedures and structures, that helps manage and condition individual perceptions of innovation in a manner that balances CA and SCA for the organization whereby supporting different types of innovation and incentivizing desired behaviors.

Chapter Summary

In conducting the literature review, several issues have emerged while attempting to frame the discussion around perception of innovation and individual's culture. The most obvious and primary challenge is the lack of a single, coherent and all inclusive definition for innovation that could be applied in all contexts. In reviewing the various definitions of innovation, there is clarity that the definition of innovation needs to be context driven and hence the research conducted also needs to be context driven. The findings of various authors organization (Baregheh et al., 2009; Denning, 2012; Kaasa & Vadi, 2008; Leavy, 2011; Roberts, 2007; Williams & McGuire, 2010), helps clarify that the definition of innovation includes a multi-step or multi-phase process, and one that involves distinct and differentiable roles for the individual and the organization. The research conducted helps clarify and decouple various parts and components of the innovation process and also sheds clarity to the roles of the individual and the organization. In addition to the aforementioned definitions and clarifications, and additionally defining innovation to be new, improved or change, in association with the various degrees of influence on perception of innovation from culture, gender, job function, and skills,

indicates the existence of a continuum for innovation that is affected by these factors change (Zhuang et al., 1999, Baregheh et al., 2009; Shin et al., 2012; Caraballo & McLaughlin, 2012).

Nations are diverse and the culture at the national level is rarely singular (Cohen, 2007). Thus, describing innovation at the national level using culture may not be fitting as there is not a single best culture that would drive successful innovation. Although, a critical role persists for the organization to play at the organizational culture level to bring about the cohesiveness between socializing and coordination of innovation activities that deliver on innovative outcomes. Organizations are made up of people and when we speak in the context of the organizations culture we are speaking in the context of a collective perception, belief and collective culture of all the people that make up such an organization over a period. Hence, perception begin at the individual level and form or collectively form the organizational culture over time. The ways of dealing with uncertainty, risks and challenges that the organization faces over time, all help shape the culture of the organization that start with perceptions at the individual level. Ultimately the lessons learnt become the way of doing business and hence become habit forming and further solidifying the culture. The individual's perceptions are expressed in a non-tangible manner via speech, actions, and behaviors and in a way accept-reject decisions. The individual's behaviors and actions as learned by being exposed to certain challenges and environments help with drawing the dichotomy and differentiating between cultures (Hofstede, 1980; Williams & McGuire, 2010). The adage, birds of a feather flock together, or in this case, socialization among like-minded individuals, who share the same perception, collectively demonstrates the perceptions of the group, while coming together and solving problems as a team, demonstrates coordination and shows the collective efforts of individuals in fulfilling the organizations strategy.

In realizing the benefits of innovation, the crucial task is to align the perceptions and efforts of the individual with those of the organization. Searching for solutions, boundary spanning and executing on solution, knowledge management are key to aligning perceptions and connecting the perceptions between the individual and the organization. Innovation that are new or improved can leverage explicit knowledge and do not find an extensive need for external or internal coordination. As opposed to radical change innovations that rely heavily on tacit knowledge that is gained via boundary spanning from a diverse talent and knowledgebase. When comparing new and improved innovations, with radical and change innovations, the risk tolerance amongst the perception of innovation at the individual and organizational level need to be high for radical innovations, while a low risk tolerance would suffice for new and improved innovations. Radical innovations that disrupt an industry, or change and shift cultures are of high risk nature and have many unknowns associated with such change. Context has a relationship with innovation in that perception of innovation by the individual level is defined by the individual while measuring risk, and the definition of risk in turn is dependent on the context (Euchner, 2011).

All cultures share the commonality of risk and uncertainty, however, what differentiates the cultures are the perceptions and variations in the manner in which risk and uncertainty are managed by respective cultures (Li, Griffin, Yue, & Zhao, 2013). Certain factors such as resource availability, severity of the risk, stability and possible duration of instability, factor into how cultures manage risk and uncertainty. Over a period, culture develop mechanisms and tactics to dealing with the unknown and risks by switching between traditional solutions driven by explicit knowledge and innovative solutions driven by tacit knowledge (Kenny & Reedy, 2006; Wilson & Doz, 2011). Hence, stable environments lean towards low-risk new and improve

innovation driven by traditional approaches, while unstable environments lean towards high-risk, change innovations and disruptive approaches. The approaches are not exclusive and form the ends of the continuum, opposite to one another and distinctly differentiated by their characteristics in Figure 4.

Age (generational cohort), gender and job function are all factors and categories that introduce variations in perception of innovation and variations in the types of innovation at the individual level (Caraballo & McLaughlin, 2012; Zhuang, 1995; Zhuang et al., 1999). Each generation cohort exhibits similar characteristic that are partially dependent on external factors. Each culture has its own distinct set of social conventions that apply towards the environment and experiences for each gender. The job that the individual is in requires a certain set of skills to, first be in the job and second to actually perform the job, thus making the job functional dictate the skill-sets needed. Although, the extent to which, job functions influence perception of innovation is unclear, as the motivation to perform the job varies and is driven by the individual's need or purpose. Hence, the individual's perspective clarifies the role for the individual to better understand perspectives and perception of innovation, which would help drive the individual's career, and selection of industry, while the role of the organization is clarified towards establishing and implementing policies, procedures and structures that allow for improvement of human resources and ensuring that such policies, procedures and structures are in alignment with the overall strategy at the organizational level.

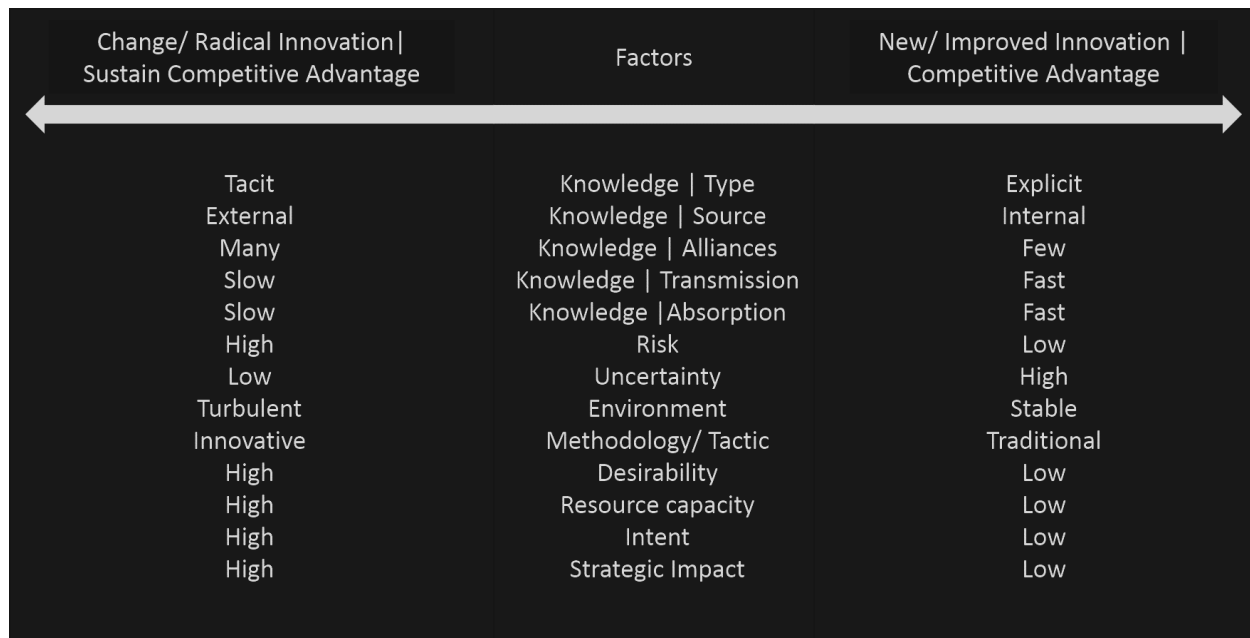


Figure 4. Innovation Continuum.

New and improved innovation rely on similar factors from an influence standpoint, especially in relation with change/ radical innovations. There is a degree of variation of the factors in terms of new and improved innovation. If broken into three categories, new innovation would be somewhere in the middle of the two extremes, change and improved. None of these would suffice as a requirement as innovation remains contextual.

CHAPTER 3. METHODOLOGY

Introduction

This research focused on furthering the comprehension of perception of innovation in Indian Americans, while identifying and correlating the relationships between gender, age, and job function association and perception of innovation. In addition, identifying the consistency of results as compared to previous studies. The research methodology used to collect and analyze data will be discussed in this chapter along with a discussion of the survey instrument used for this research. This chapter also discusses the research design/ methodology, population, sample, variables, setting, data analysis techniques, validity and reliability of the survey instrument and ethical considerations. The research was quantitative, exploratory, and non-experimental, using an updated survey instrument for data collection, following the design of Caraballo and McLaughlin's (2012) research. Given the exploratory nature of the research, according to Creswell (2009), this type of data collection was suitable and appropriate for such research that involves generalizing results from a sample to a larger population.

The research conducted by Caraballo and McLaughlin (2012) leveraged the preliminary work conducted by Zhuang (1995) and Zhuang et al. (1999), which used a 10-item survey to measure individuals perception of innovation in medium size businesses in England. Because the studies conducted by Zhuang (1995) and Zhuang et al. (1999) resulted in mixed results, the authors decided not to further pursue their initial exploratory work. Focusing on a different population and selecting a different data collection methodology, Caraballo and McLaughlin

(2012) were able to extend the initial work conducted by Zhuang (1995) and Zhuang et al. (1999), and developed statistically significant results.

Because of the broad and multiple definitions and wide applicability of innovation, there is a lack of clear and specific understanding of innovation along with multiple views of how innovation may be perceived (Baregheh et al., 2009). Further, at the individual level, there is little research on the impact of culture, which also drives perceptions of innovation at the individual level (Blodgett, Bakir, & Rose, 2008; Caraballo & McLaughlin, 2012). This research addressed a gap in the literature in terms of how innovation is perceived by individuals.

Research Design

This research extended and built upon the exploratory research work conducted by Caraballo and McLaughlin (2012), and the research was quantitative, exploratory, and non-experimental using a web-based survey for data collection (Creswell, 2009). Creswell (2009) noted that surveys are appropriate data collection instruments to generalize from a sample of a target population and infer about specific characteristics, attitudes or behaviors of the target population. The intent of this study was to use web-based surveys for data collection purposes following the design established and used by Caraballo and McLaughlin (2012). The survey used for this research was self-administered, cross-sectional, and web-based, which helped contain costs, was sustainable, eliminated unnecessary variables, in addition to added convenience, for both the participants and the researcher. The researcher recognized that web-based survey approach has trade-offs between benefits and limitation. The benefits of the web-based approach include lower costs and convenience of data collection. The disadvantages of the web-based approach include a low response rate and the greater possibility of non-response bias. Such

limitations are consistent with the web-based survey approach and does not negate the usability of expected responses (Cooper & Schindler, 2007).

In order to confirm the validity and reliability of the results, Caraballo and McLaughlin (2012) conducted a Varimax Rotation, and Q factor analysis to examine the independent demographic variables. This research furthered the work done by Caraballo and McLaughlin (2012), and applied the same approach and techniques to the three independent demographic variables: gender, age (by generational cohort) and functional association (job function), and identify relationships with dependent variables on perceptions of innovation: improvement/ change, adoption of something that exists elsewhere, and new ideas/ something new. The dependent variables were identified as factors 1, 2, and 3, respectively, by Caraballo and McLaughlin (2012).

This research adopted and followed the research conducted by Caraballo and McLaughlin (2012), using a quantitative and exploratory approach. The primary difference between this research and the research conducted by Caraballo and McLaughlin (2012) was collection of data from a different sample that was previously unstudied.

The two research questions identified by the researcher for this study were:

1. What are the perceptions of innovation in Indian-American professionals residing in the US as they categorize/ associate innovation as new, improvement, or change?
2. What is the relationship between gender, age (generational cohort), functional association (job-function), and an individual's personal perception of innovation as they categorize/ associate innovation as new, improvement, or change?

The research addresses the management problem of increasing the comprehension of the different ways in which perception of innovation influences individuals means as they categorize

innovation to improve the innovation process (new product or service development) and to increase the effectiveness of collaboration with diverse partners and markets. A potential outcome of the study will be greater insight into how innovation is perceived by individuals, which in turn will enhance team formation, improve comprehension of perception of innovation by diverse markets, provide information and enable the process of being partners as companies build virtual organizations (Gressgård, 2011). The ultimate goal of obtaining these insights was to support and leapfrog the development of competitive advantage for organizations that are amenable to accepting and leveraging this knowledge (Amberg, Reinhardt, & Kittler, 2008)

Sample

Selection of Population

The target population for this study consisted of Indian American professionals in the United States. Several reasons exist for focusing on this demography. First and foremost, one of the primary reasons for selecting this group is because the Indian American group has not been studied in terms of perception of innovation because of the relatively small population in the United States. Secondly, as Project Impact expands globally, the insights from this research could potentially be applied in the Indian sub-continent. Thirdly, according to the 2010 US Census, while Indian Americans represent 1% of the total US population, this group showed a 69.3% increase since the 2000 census. This rate of increase is comparable to that of the Hispanic/Latino population which comprised 16.3% of the 2010 US population and was the target demography in the initial study conducted by Caraballo and McLaughlin (2012). Fourthly, according to the 2010 US census data, Indian American households have the highest household income, with a median household income of \$88,538 (2009 census), amongst all the ethnic groups in the United States. Hence, even though the absolute numbers are smaller for the

group, the population selection demonstrates exponential and significant economic impact, both nationally and internationally (<http://www.census.gov/prod/cen2010/briefs/c2010br-11.pdf>).

Sample

This research does not have a sponsoring organization. The researcher planned to advertise the study and solicit participants and responses via the Qualtrics website. Following filters were established by the researcher for this research: Indian American adults over the age of 18. The survey instrument had questions on number of years of work experience and area of work to ascertain the professional and job function/ association of respondents. The distribution of advertisements for this survey was done by Qualtrics to its existing database of participants using the established filtering criteria of Indian American adults over the age of 18, which produced a random sample of participants meeting this criteria.

The researcher made the assumption that there will be at least 200 complete responses with enough density in multiple professions and both genders, to afford for meaningful inference and comparison. The approach was considered acceptable for exploratory research, even though selecting a random sample in this manner is a deviation from the initial study conducted by Caraballo and McLaughlin (2012). The use of web-based survey was consistent with the work conducted in the prior study by Caraballo and McLaughlin (2012). The descriptor Indian Americans refers to individuals who can trace their descent to India. While there are several sub-cultures, religions and states within India, similar to Latinos and their country of origin in the study by Caraballo and McLaughlin (2012), this aspect of diversity within India from a descent perspective was beyond the scope of this study as it was exploratory in nature as was not included as part of the work by Caraballo and McLaughlin (2012) or Zhuang (1995) or Zhuang

et al. (1999). Protection of individual privacy and the ethical concerns therein are addressed in accordance with Qualtrics Privacy Protocol, which has been included under.

Sampling Methodology

Using a quantitative approach, this research was confirmatory in nature. According to Cooper and Schindler (2007), quantitative study is appropriate for confirmatory research. Following the design of Caraballo and McLaughlin (2012), a survey methodology was selected for quantitative research. In order to derive conclusions about the entire population, a sample was used, which also offered additional benefits of lower cost, greater accuracy of results, faster or speedier data collection, and availability of population elements (Cooper & Schindler, 2007). The sample frame was restricted to Indian Americans over the age of 18 who respond to the survey, which satisfied the requirement identified by Cooper and Schindler (2007) of ensuring the sample frame to consist of elements from which the sample is actually drawn and constitutes “a complete and correct list of population members only” (p. 397). The panel of participants for this sample constitutes those Indian American adults over the age of 18, who have self-selected to complete the survey. As identified by Cooper and Schindler (2007), the self-selection aspect meets the criteria for an acceptable panel for survey research. The population element in this research was the individual, which is the same as in the research conducted by Caraballo and McLaughlin (2012).

According to Cooper and Schindler (2007), probability sample procedures are appropriate for confirmatory research and this research used probability procedure and incorporated random sampling. By the very nature, a random sample enables every individual to have an equal opportunity to participate (Cooper & Schindler, 2007). Confirmatory studies seek to test one or more hypotheses pertinent to the relationship between dependent and independent

variables. Thus making all of these approaches to be mutually supportive and appropriate for the methodological approach in this research.

Setting

The research was conducted using a web-based survey involving Indian American adults over the age of 18 who self-select to participate in this study. The research conducted by Caraballo and McLaughlin (2012) focuses the survey on members of one specific organization, a web-based survey approach is deemed appropriate for collecting the type of data most useful for such research. Qualtrics was provided with electronic versions of the welcome letter, instructions, informed consent authorization, and link to the survey on the internet by the researcher. The survey was advertised by Qualtrics to their database of members and Qualtrics distributed informed consent and survey to those who respond positively to participate. There was no direct or indirect access between the researcher and individual member's identifiable information. The ability to self-select was enabled for all participants of the survey, along with the ability to opt-out of the survey, should the participants choose to do so. Incomplete surveys were not be included as part of the study.

Instrumentation

For this research, the researcher selected the modified survey items from Caraballo and McLaughlin's (2012) research, which were modified from Zhuang's (1995) survey items that describe how an individual perceives innovation. Following the work of Caraballo and McLaughlin, hypothesis testing was used to determine if there was a difference between the mean response for each statement as well as a test of homogeneity of variance of the responses to determine variability of the responses. A survey (Appendix B) consisting of a total of 20 questions was used to collect data, while the questions were spread out over three sections.

Zhuang used a 10-item survey to data on how individuals perceived innovation and as to how individuals characterized innovation in new, improvement or adoption. Zhuang et al. (1999) combined two surveys to avoid repetitive questionnaires to respondents, but this made the questionnaire too long and resulted in costly incomplete responses. Caraballo and McLaughlin choose Zhuang's survey to collect and analyze information on how individuals understood innovation as applied to a specific demography. In examining the survey statements and responses, Caraballo and McLaughlin used the three demographic characteristics of gender, job function and age and identified by generation. Eight of the twenty questions addressed personal demographic information, such as gender, age, geographical part of the country in which the respondent resides, highest level of education, length of time employed in the current position and with the current employer, general job classification, industry in which employed, and if the respondent is in a technical or a staff position.

Zhuang (1995) did not provide the Cronbach's alpha for the 10-question survey, so an important measure of internal reliability was not reported. Zhuang et al. (1999) also failed to report the Cronbach's alpha for their innovation survey. In the present study, the Cronbach's alpha for the 10 survey items pertaining to perceived characteristics of innovation was measured and reported. A Cronbach's alpha greater than .7 will be considered to confer a sufficient level of internal reliability on the study, in keeping with the criteria specified by Creswell (2009). The Cronbach's alpha for the survey instrument was calculated to be 0.883 (Appendix G).

Twelve of the twenty questions required forced choices of Yes/ No addressing the individual and their respective work environment. Finally, the survey concludes with questions that address how respondents comprehend the different aspects of innovation: six of these questions were constructed on a five point Likert Scale ranging from Strongly Agree to Strongly

Disagree; and the rest were forced choice questions. Appendix B has a copy of the survey. Permission had been given by Zhuang, Caraballo and McLaughlin to use their survey, and have documented the validity and reliability of the instrument (Caraballo & McLaughlin, 2012; Zhuang, 1995; Zhuang et al., 1999).

Table 1

Variables in the Study

Independent	Dependent
Gender	New
Age (Generational Cohort)	Improved
Functional Association	Change

Data Collection

Qualtrics.com hosted the survey electronically. Individuals who selected to participate, were presented on the first screen an explanation on the introductory email and information on the purpose of the survey, its length, and guidelines for contacting the researcher to receive a copy of the results. Participants were assured of their confidentiality and of safekeeping of all data collected. Participants/ respondents were also informed that completion of the survey would equate to their informed consent to participate in the study. Participants were asked certain demographic questions as part of the survey, however, they were informed that no personal information identifying them uniquely with their name, address, telephone, or organizational membership would be collected, and that their responses were coded to prevent identification. In addition to the above assurances to protect individual identification, participants were also informed that any and all data reported used for the study will be aggregated to ensure

confidentiality of the individual responses. Participants were asked to indicate that they have read the instructions on the first screen of the survey and clicking on the “Next” button indicates their acknowledgement. Forced answering was required only on the entry page before moving to the next screen. A maximum of two weeks was provided to participants to during which they can participate and respond to the survey. Follow-up communications by Qualtrics were sent to prompt those who have not responded to the survey, after the first invitation email has been sent.

Data Analysis

The researcher had identified the following null and alternate hypotheses in support of the questions:

Question 1:

H01. There is no mean difference between the ten survey statements.

Ha1. There is a mean difference between the ten survey statements.

H02. There are no differences in the variances between the 10 survey statement responses.

Ha2. There is a difference between the variances for the 10 survey statement responses.

Question 2:

HO3. Gender does not influence perception of innovation.

Ha3. Gender does influence perceptions of innovation.

HO4. Generational cohort does not influence perceptions of innovation.

Ha4. Generational cohort does influence perceptions of innovation.

HO5. Functional association does not influence perceptions of innovation.

Ha5. Functional association does influence perceptions of innovation.

This study also incorporates the additional hypotheses developed by Caraballo and McLaughlin's (2012). Null and alternative hypotheses are as follows:

H06. Mean of Factors 1, 2, and 3 are not statistically different.

Ha6. At least one Factor mean is different.

The data analysis approach in this study followed the same approach used by Caraballo and McLaughlin (2012). Survey items from Zhuang (1995) were used, which described how individuals perceive innovation. As noted by Caraballo and McLaughlin (2012, p.555), "hypothesis testing will determine if there is a difference between the mean responses for each statement as well as a test of homogeneity of variance of the responses to determine variability of the responses". The author proposed to use an Analysis of Variance (ANOVA) technique at the 95% confidence level to test the 10-survey response means, following the Caraballo and McLaughlin (2012) approach. IBM SPSS 22 software was used to perform the statistical analysis.

The author also tested the homogeneity of variances at the 95% confidence level to verify if individual responses vary when assessing different elements (meanings) of innovation. Outliers will have an impact on the homogeneity of variance. The research conducted by Caraballo and McLaughlin (2012) followed best practices, and no outliers were detected in the study. Because this study focuses on a single group of specific national descent, transferability of results would be inappropriate regardless of sample size.

"Factor analysis condenses (summarizes) information contained in number of original variables into a smaller set of new, composite dimensions or factors with a minimum loss of information" (Hair, Anderson, Tatham and Black, 1995, p. 368). This study used factor analysis to determine the variability or intactness of the Zhuang (1995, 1999) and Caraballo and

McLaughlin (2012) three dimensions/ factors (gender, age and functional association) of an individual's perception of innovation. In order to ascertain which concepts (constructs) emerge from the survey statements and if the concepts are independent of one another, the researcher used factor analysis. Factor analysis helped determine the variability in the three factors.

Validity and Reliability

The initial instrument was piloted and used by Zhuang (1995), which underwent revisions that added and deleted certain questions subsequently. The modified instrument was used by Zhuang et al. (1999) for their research; however, neither Zhuang nor Zhuang et al. reported the Cronbach's alpha for their respective studies. Caraballo and McLaughlin (2012) used the same instrument, from Zhuang et al. (1999) and found an alpha reliability estimate of 0.675, which shows small amount of inconsistency in data, but proves acceptability as a measure of overall reliability of the survey. Thus, Caraballo and McLaughlin used the same instrument and did not find any issues with reliability or validity of the instrument for their survey. The researcher used the modified version of the survey instrument from Project Impact, which yielded Chronbach alpha of 0.883, hence improving the validity and reliability of the instrument.

Ethical Considerations

Potential risks and benefits discussed by Creswell (2009) in terms of data collection, analysis, interpretation, writing, and dissemination of the research have been incorporated into the study. From an ethical standpoint, participants did not benefit directly unless such participants request a copy of this completed study. Upon completion of the survey, participants might benefit indirectly in terms of enhanced comprehension of how perceptions of innovation differ, depending on an increased awareness of this matter.

Ethical standards were maintained through Capella University IRB approval and the study did not involve more than a minimal level of risk for the participants. None of the individuals were pressured via influence to self-select to participate in the taking the survey by the researcher. All participants were made aware that that their participation in the study was voluntary and that they have the ability to opt-out at any time. At the beginning of the survey, participants were asked to indicate their informed consent. Measures such as, not collecting personal identification information (names, email addresses, phone numbers, or addresses, etc.) were followed to ensure protection of participant confidentiality. Instead, each participant responding to the survey will be assigned a code. Further, all responses were matched to the assigned participant code only. The Qualtrics.com website collected all data and in accordance with the privacy and security policies of Qualtrics.com, Qualtrics.com will destroy all copies of the data and any analysis, after transferring these data and analysis to the researcher. All data are electronically safeguarded by the researcher via mandatory password requirement to access data. Using password protection, the researcher securely stored data on his personal computer, and in a removable media in a locked file cabinet, for the requisite period of seven years, at the end of which, the data shall be destroyed and erased from all electronic media in which data may have been stored, by electronic sanitization on the removable media as well as on the personal computer.

The study did not include protected groups defined by the Belmont Report (The Belmont Report HHS.gov). The focus of the study on a demography with a specific ethnicity and national descent, may be interpreted as a violation of the of the provisions of the Belmont Report, however, the researcher believes that measures such as self-selection to participate and the ability to opt-out at any time will provide adequate protection and means that will prevent harm to any

participant. While the possibility of one or more pregnant woman participating in the survey may exist; this was a self-selection that is not controlled by the researcher. Moreover, the approach described for this study replicates the approach used in previous study of a single ethnic group without incidence of harm. The researcher believes that these provisions, minimal level of risk to the participants, and the potential benefit of increased comprehension of this subject argue for approval of this study.

CHAPTER 4. RESULTS

Introduction: Advancing the Research

Building on the seminal and scholarly research conducted on perspectives of innovation by Zhuang (1995), Zhuang, Williamson, and Carter (1999) and, Caraballo and McLaughlin (2012), the research focuses on the specific demography of Indian Americans (Asian-Indian decent) in the United States. Using the definition of individual innovation from Zhuang's research, McAdam and McClelland (2002) extended the research with a focus on creativity generation at the individual and team level. McAdam and McClelland's (2002) research focused on the three elements of creativity: expertise, task-motivation, and creative thinking with specific examination on the source of ideas. However, research on the source of ideas remains limited. Hence, Project Impact, a non-profit organization dedicated to innovation research, leveraged the Zhuang's revised instrument (1995), to determine "if a homogenous group of respondents employed by various companies in the IT industry within the United States, perceived a similar concept of innovation using the Zhuang (1995; 1999) three-element perspective" (Caraballo & McLaughlin, 2012, p.555).

Advancing the Research

With permission from Project Impact, this research leverages the revised instrument to advance the research and determining if a homogenous group of respondents (Indian Americans), from various companies in the United States perceive the similar concept of innovation using the three element perspectives of expertise, creative-thinking, and task motivation. The validity and reliability of the instrument has been documented by Caraballo and

McLaughlin (2012) and Zhuang (1999). The research identifies potential correlations and relationships between Indian-Americans perceive innovation at the individual level relative to variables such as gender, age, functional association, and education). The results of this research could benefit leaders, decision makers and organizations in being better prepared during recruiting, strategic planning, building high-performance teams and structuring the organizations for successful innovation. The research could also help organizations with an understanding of cultural perceptions, thereby providing information when attempting to penetrate or broaden market reach for the Indian consumer. The remainder of this chapter will offer a description of the sample, summary and detailed analysis of the research results, and summarizing the final conclusions regarding the research findings.

Description of Sample

As discussed in Chapter 3, the research intended to gather responses from 200 respondents, using the Qualtrics platform that ran the survey in a voluntary and anonymous manner for the Indian-American demography. The Qualtrics panel ensured responses were only collected from Indian-Americans (Asian-Indian decent), living in the United States. The survey collected data for 8 days from March 20, 2014 to March 28, 2014, during which period a total of 200 usable responses were received. The primary factor for choosing Indian-Americans is the researcher's familiarity with the Indian demography. Table 2 provides the demographic distribution of the respondent's profiles.

Table 2

Distribution of Demographic Profile of Participants

Demographic Information	Frequency	Percent
Gender		
Male	101	51%
Female	99	49%
Age (Generational Cohort)		
Born before 1947	3	1%
Born between 1947 and 1963	8	4%
Born between 1964 and 1979	56	28%
Born after 1980	133	67%
Technical Role		
Yes	93	47%
No	106	53%
Education		
Doctoral degree	18	9%
Master's degree	77	38%
Bachelor's degree	80	40%
Two-year degree	10	5%
Certificate	6	3%
High school or equivalent	10	5%

Findings

Research Question 1: What are the perceptions of innovation in Indian-American professionals residing in the US as they categorize/ associate innovation as new, improvement, or change?

Testing the following sets of hypotheses answers the research question. The first set of hypotheses draws a comparison of the mean responses from the 10 survey questions that respondents provided answers to. The second set of hypotheses determines whether there is a significant difference between the variances for the responses from the 10 survey questions.

H01 (Null Hypotheses): There is no mean difference between the ten survey statements.

Ha1 (Alternate Hypotheses): There is a mean difference between the ten survey statements.

H02 (Null Hypotheses): There are no differences in the variances between the 10 survey statement responses.

Ha2 (Alternate Hypotheses): There is a difference between the variances for the 10 survey statement responses.

Mean Comparison. Statistical analysis were conducted using IBM SPSS Version 22 software. To test the set of hypotheses, the 10-survey response means, the Analysis of Variance (ANOVA) technique at 95% confidence level was used. The one-way ANOVA between subjects compared the effect of the concepts or definitions of innovation in terms of how Indian American professionals perceived innovation. A p -value of $p < 0.05$ presents evidence to reject the null hypotheses, thereby testing that there is a mean difference between the 10 survey statements. Table 3 shows that the statements tested to be statistically significant. This indicates that the responses of the 200 participants to the variations in the definitions of innovation presented in the survey are relatively different to each other. Table 3 presents the results from the ANOVA analysis.

Table 3

ANOVA Analysis Results

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	36.844	9	4.094	5.72	0.000
Within Groups	1423.535	1990	0.715		
Total	1460.380	1999			

S = 0.8458 R-Sq = 2.52% R-Sq(adj) = 2.08%

Homogeneity of Variances. A statistically significant test of homogeneity of variances

verifies that individual responses from Indian-Americans vary when assessing different definitions of innovation. Homogeneity of the data was tested using the Levene Test of Equality of Variances. As indicated in Table 4, the analysis conducted homogeneity of variance test at the 95% confidence level. The results of which revealed that variances are homogeneous, no differences. With a p -value of $p > 0.05$, there is evidence that variances are homogenous along with the error value of the dependent variable being equal across groups. Hence, the null hypotheses may be accepted that there are no differences in the variances between the 10 survey statements.

Table 4

Levene's Test of Equality of Error Variances

	F	df1	df2	Sig.
In your opinion, innovation can be defined as-a. Inventing something entirely new	.044	1	198	.833
In your opinion, innovation can be defined as-b. Generating new ideas	.182	1	198	.670
In your opinion, innovation can be defined as-c. Improving something that already exists	.316	1	198	.575
In your opinion, innovation can be defined as-d. Enhancing the desirability of something	.416	1	198	.520
In your opinion, innovation can be defined as-e. "Thinking out of the box"	2.913	1	198	.089
In your opinion, innovation can be defined as-f. Taking something that already exists and making it better	.014	1	198	.907
In your opinion, innovation can be defined as-g. Changing the technology by making the previous version obsolete	.000	1	198	.991
In your opinion, innovation can be defined as-h. Adopting something that has been tried elsewhere	.353	1	198	.553
In your opinion, innovation can be defined as-i. Using a different perspective	.497	1	198	.482
In your opinion, innovation can be defined as-j. Introducing change to make something better	.255	1	198	.614

Research Question 2: What is the relationship between gender, age (generational cohort), functional association (job-function), and an individual's personal perception of innovation as they categorize/ associate innovation as new, improvement, or change?

To answer the research question, the researcher tested the following set of hypotheses using Multivariate ANOVA tests using IBM SPSS version 22. The assumptions of normality and assumed independence are not tested but assumed. The first set of hypotheses refers to the comparison of the mean responses from the 10 survey statement responses according to the gender while the second set of hypotheses refers to determining whether there is a significant difference between the responses of the generational cohort to the 10 survey statement responses. Lastly, the third set of hypotheses refers to the comparison of the mean responses from the 10 survey statement responses according to the functional association of the participants.

H03 (Null Hypotheses): Gender does not influence perception of innovation.

Ha3 (Alternate Hypotheses): Gender does influence perceptions of innovation.

Mean Comparison. Multivariate tests on the responses from 200 Indian-Americans yielded in the results shown in Table 5. The sample of Indian-American respondents included 101 (51%) males and 99 (49%) females. As noted in Table 5, p -value of $p < 0.05$ presents evidence to reject the null hypotheses, thereby testing that gender does influence perception of innovation in Indian Americans. This indicates that the participant being a male or female influenced respective perception of innovation between new, change or improved.

Table 5 *Multivariate Tests*

Effect	Test	Value	F	Hypothesis		
				Df	Error df	Sig.
Gender	Pillai's Trace	0.127	2.574	10.00	177.00	0.006
	Wilk's Lambda	0.873	2.574	10.00	177.00	0.006
	Hotelling's Trace	0.145	2.574	10.00	177.00	0.006
	Roy's Largest Root	0.145	2.574	10.00	177.00	0.006

Age	Pillai's Trace	0.148	0.932	30.00	537.00	0.573
(Generational Cohort)	Wilk's Lambda	0.857	0.937	30.00	520.20	0.565
	Hotelling's Trace	0.161	0.942	30.00	527.00	0.558
	Roy's Largest Root	0.101	1.807	10.00	179.00	0.062
Functional association	Pillai's Trace	0.046	0.416	20.00	356.00	0.989
	Wilk's Lambda	0.954	0.417	20.00	354.00	0.988
	Hotelling's Trace	0.048	0.419	20.00	352.00	0.988
	Roy's Largest Root	0.044	0.783	10.00	178.00	0.645
Gender	Pillai's Trace	0.309	2.059	30.00	537.00	0.001
Age	Wilk's Lambda	0.714	2.105	30.00	520.21	0.001
	Hotelling's Trace	0.367	2.150	30.00	527.00	0.000
	Roy's Largest Root	0.254	4.555	10.00	179.00	0.000

HO4 (Null Hypotheses): Generational cohort does not influence perceptions of innovation.

Ha4 (Alternate Hypotheses): Generational cohort does influence perceptions of innovation.

Mean Comparison. Multivariate tests on the responses from 200 Indian-Americans, which as noted in Table 2 included respondents categorized into four generational cohorts: born before 1947, born between 1947 and 1967, born between 1968 and 1979, and born after 1980. Those born after 1980 may be categorized as generation X or millennials. The analysis of the responses, conducted using Multivariate ANOVA tests using SPSS version 22 indicated in Table 5 show a p -value of $p > 0.05$, thus providing evidence that the null hypotheses be accepted. This shows that age or generational cohort, does not influence the perception of innovation. Respondents in the 4 generational cohorts did perceive innovation differently when categorizing innovation into new or change/ improvement.

In analyzing the combination of the gender and age in regards to perception of innovation, revealed that the combination has a $p < 0.05$, indicating that there is no influence on the perception of innovation.

HO5 (Null Hypotheses): Functional association does not influence perceptions of innovation.

Ha5 (Alternate Hypotheses): Functional association does influence perceptions of innovation.

Mean Comparison. Functional association was categorized as technical versus non-technical. As noted in Table 2, out of the 200 respondents, 93 were technical and 106 were non-technical. Multivariate tests, shown in Table 5 yielded a p -value of $p > 0.05$, thus indicating evidence from the tests that the null hypotheses be accepted. Thereby, showing that functional association does not influence the perception of innovation.

From the results Multivariate tests of the survey for 200 Indian Americans, gender seems influence the perception of innovation, while age and functional association do not seem to influence the perception of innovation. Table 5 presents the results from the multivariate ANOVA analysis that shows the interaction between gender and age, is statistically significant with $p < 0.05$. Hence, indicating that the interaction of gender and age does have an effect the mean response of constructs 1, 2 and 3.

This study also incorporates the additional hypotheses developed by Caraballo and McLaughlin's (2012). Null and alternative hypotheses are as follows:

H06 (Null Hypotheses): Mean of Factors 1, 2, and 3 are not statistically different.

Ha6 (Alternate Hypotheses): At least one Factor mean is different.

Caraballo and McLaughlin (2012) formulated three fundamental constructs or dimensions where people associate the definition of innovation: new, adopting, and improve/change. For the purpose of this study, the researcher uses the same, following the methodology of Caraballo and McLaughlin (2012): the ten survey statements are categorized

into these three constructs. Table 6 presents the allocation of each statement to one of the three constructs of innovation.

Table 6

Allocation of each statement to the construct of innovation

	Construct 1	Construct 2	Construct 3
Statements	c, d, f	g, h, i, j	a, b, e
Naming convention	Improve	Change	New

Minitab 14 was used to conduct a One-way ANOVA technique that tested the additional hypotheses developed by Caraballo and McLaughlin (2012). The additional null and alternative hypotheses are as follows:

H06 (Null hypothesis): Mean of Factors 1, 2, and 3 are not statistically different.

Ha6 (Alternative Hypothesis): At least one Factor mean is different.

The One-way ANOVA tested that there is a statistical difference. The test showed that there is evidence to reject the null hypothesis ($p < 0.05$) and accept the alternate hypotheses that at least one factor mean is different. Hence, indicating that the mean of factors 1 and 2 are statistically different and that gender, generational cohort, and functional association do not have an effect on the mean response of the constructs.

Table 7

One-way ANOVA Analysis Results

Source	df	Sum of Squares	Mean Square	F	Sig.
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Factor	1	5.192	5.192	12.43	0.000
Error	398	166.224	0.418		
Total	399	171.416			

S = 0.6463 R-Sq = 3.03% R-Sq(adj) = 2.79%

Factor Analysis

In conducting factor analysis of the data from the 10 survey statements from the 200 responses indicated that the sample did not differentiate the perception of innovation into three categories of new, change and improvement. Instead there were only two components, which were new and change/ improve. Thus, indicating that respondents in this sample identified the dichotomy in the perception of innovation only into two categories. However, adding the factor of education, highlighted the third component. Varimax rotation indicated that that higher the level of education (Masters or higher) led to the respondents being able to differentiate perception of innovation into three categories of new, improve and change as opposed to the previously note two categories of new and improve/change. Hence respondents with higher levels of education were able to differentiate between change and improvement, when it came ot perception of innovation. In conclusion, factor analysis highlighted that even though gender and age have a similar profile, there are statistical differences in gender and age (Appendix C). The addition of the factor called education resulted in differences between the groups in how they classified innovation.

Summary

The purpose of this research was to examine how Indian Americans perceived innovation, and identify the relationships and correlation between the perception of innovation and gender, age, or functional association within the homogenous group of Indian Americans. In conducting the research, responses were collected from an Indian-American demography using

the Qualtrics survey tool, which was used to develop the survey that was a modified version of the survey used by Zhuang (1995) and was modified by Caraballo and McLaughlin (2012). Analysis was conducted on a total of 200 responses using the IBM SPSS 22 platform and techniques such as ANOVA were applied to test the six hypotheses that answered the two research questions. The results of the analysis revealed that there is a mean difference between the 10 survey statements and there is no difference in the variance between the 10 survey statements. From the analysis and tests of the 200 responses from Indian Americans, there is evidence that gender does influence perception of innovation. However, age and functional association (technical and non-technical) have not influence on the perception of innovation. Further factor analysis also determined that in definition or drawing a dichotomy in the three categories of new, improve and change, respondents only showed to identify innovation with new or change/ improve. With the exception of education as a factor and at the Master's or higher level, respondents perceived innovation to fall into two categories and did not draw any significant difference between change and improvement. The test of equality proved to be statistically significant. The multivariate analysis revealed that at least one factor has a different mean among the three constructs. Chapter 5 addresses the discussion on findings, conclusion and recommendations for this research.

CHAPTER 5. DISCUSSION, IMPLICATIONS, RECOMMENDATIONS

Discussion

The research focused on the management problem of understanding how innovation is perceived within a homogenous group of individual, specifically understanding how Indian Americans perceived innovation and how did Indian Americans associate innovation with new, improvement or change. The research broadens the current research conducted by Caraballo and McLaughlin (2012) by focusing on a new demography in the United States. The literature review covered a review of literature that sheds light on innovation and culture at level of the individual, team and country. Although, there is significant body of literature on cultural impact on the perception of innovation at the organizational and country level, there is also a level of contradiction. Also, there is less research conducted on the perception of innovation within a specific cultural demography at the individual level. Inconsistencies in the definition and subject of innovation and its perception continue to persists, which is a continuation of findings from the literature review. However, the constructs of new, improved and change identified by Baregheh et al. (2009) and tested by Caraballo and McLaughlin (2012) provide a sustainable framework to continue to broaden the study of the subject. Leveraging these constructs Caraballo and McLaughlin (2012) were able to identify that perception of innovation within a homogenous group could also vary, while the factors of age, cohort and job functions did not influence the perception of innovation in terms of new, improved or change. Chapter 5 provides an overview of this research conducted within the homogenous cohort of Indian Americans and discusses implications, limitations and recommendations for future research.

Overview

Preliminary results from the research identified that there is a difference in how Indian Americans perceive innovation. In general, participants exhibited an understanding of innovation as a multi-dimensional construct. Perception of innovation varied in respondents when the dimension of education was introduced as part of the analysis. Respondents who had a Masters or Doctoral education identified and perceived innovation into the three constructs, however, respondents with a Bachelors or lower degree identified innovation into two areas of new and improve/ change. Though the findings from the hypotheses tests were similar to the findings of Caraballo and McLaughlin (2012), statistical differences were seen in comparing gender and in interaction between gender, age and functional association. Both, this research and the research conducted by Caraballo and McLaughlin (2012) focused on a homogeneous demography at the individual level. This research focused on the Indian American demography across the United States and across industries, while the research conducted by Caraballo and McLaughlin focused on Hispanic Americans in the Information Technology (IT) industry. Both studies indicated that individuals from two different cultures exhibited and understanding of innovation as a multi-dimensional construct producing similar results. Hence, a safe assumption could be made that perceptual differences are not exclusive to one culture or ethnicity. However, further research is needed to be done to confirm the broader assumption.

The research confirmed the previous work of Caraballo and McLaughlin (2012) in that members of a homogeneous group from the same culture did perceive innovation in a different manner. From a cultural standpoint Indian approach innovation as new or change/ improve, and change /improve are seen as “jugaad”, a risk driven system that optimizes processes (Birchnell,

2011). Prior research at a national level suggested behavioral shaping to be an outcome of shared problem solving and sharing of experiencing, especially in times of uncertainty and risk, coupled with economic growth and national culture (Franke, Hofstede, & Bond, 1991; Hofstede & Bond, 1988; Kaasa & Vadi, 2008). Understanding this is important in a global market with global teams that comprise members from diverse cultures and hence making a global team a collection of perceptions influenced by diverse national cultures. Finally performance of the team directly impacts performance of the overall organization as well.

Factor analysis using Multivariate (ANOVA) tests at the 95% confidence level showed that there are statistical differences in gender and age, although both have a similar profile. Gender showed statistical differences but age and functional association did not show any statistical differences. Gender does not influence perception of innovation, while age and functional association did influence perception of innovation. Further factor analysis showed evidence that the interaction gender and functional association (technical or non-technical) does influence the perception of innovation. For instance, a certain function, product or service might seem technical to Indian American males but may not seem technical to Indian American females. This is important to know from a team building, project management, marketing and sales perspective. Depending on the teams and assignment of the job function, there could be flexibility in task assignment, hiring and developing marketing strategies. The results were consistent with Caraballo and McLaughlin (2012).

Factor analysis also pointed out that there were only two components instead of three. Component 1 was “new” and component 2 was “improve/ change”. This indicates that Indian American respondents did not draw a significant dichotomy in “change” and “improve” from a perception of innovation standpoint. The responses indicated that participants were able to

clearly define and differentiate innovation when it was new versus change/ improve. This was consistent with Caraballo and McLaughlin (2012). However, further factor analysis using the dimension of education indicated that respondents with a degree of Masters or Doctorate were in fact able to define or perceive innovation into the three constructs of “new”, “change” and “improve”. Respondents with a Bachelor or two-year degrees were unable to identify this differentiation. This is an important finding, especially when formulating or managing teams and project groups that might have a combination of Indian Americans with varying educational degrees. In addition, as part of communications, or when presenting an idea as an innovative idea or innovative process or innovative product, Indian Americans with a Masters or a Doctorate have greater potential to differentiate between innovation that may be a change versus improvement.

Culture, creativity and control (autonomy) could shape behaviors, beliefs via an incentivization or penalization mechanism that might result in innovative outcomes driven by the vectors of openness and creativity fostered or supported within the organization (Kaasa & Vadi, 2008; Kenny & Reedy, 2006). The distinctions and perceptions are crucial from the standpoint marketing and sales. For an organization, tailoring the message, product and service to the needs of the customer is important for sustain competitive advantage in the market. Understanding the perception and culture driven behaviors of the customer afford the company to offer respective service or product in a manner that might be both acceptable and differentiable from the customer’s standpoint. Understanding perception of innovation could not only help with retention of customers but also assist in attracting new customers.

The same concept and approach applies to teams and groups that may consist of Indian-Americans. Aligning individuals and convincing all proponents is required to achieve traction

and collaborative support for accomplishing innovation (Shane, 1994). A better understanding of the perception of innovation as perceived by team members affords project managers the ability effectively and efficiently communicate and execute on projects. As this study suggests, perception of innovation differs even within the homogenous culture at the individual level and hence, comprehension of those perceptual difference could afford organizations to solve problems and help collaborate on new, improvement or changes to products or services offered in a collaborative manner, while overcoming organization inertia to change. Organizational leaders would be able to establish the necessary structure, mechanisms and incentives to drive and motivate innovation within the organization in a desired direction that is strategic and customer focused. Keeping internal operations in sync with external customer driven demand needs to be part of the ideation process. Such a connection would avoid gaps that could drive teams in a direction not suitable from the company's or customer's desired direction, although such a change might seem innovative. And hence the need to ensure a structured approach to ideation, where there i idea and problem/ gap handshake that ensured there is pairing between internal ideas and external needs from customers. Understanding of perceptual differences affords the organizations leadership to align teams, functional assignments and most importantly, personnel or individuals in appropriate and close alignment with the strategic goals of the organization. Via the individual's behaviors and actions, the organization is impacted by every individual who works at the organization (Gee & Gee, 2011).

Implications

The cultural variations and understanding highlights and help drive strategy for innovative delivery of products and services that are well received and appropriately perceived by the Indian American demography. Further, an understanding from a cultural stand point on the definition

and perception of innovation by Indian Americans, can help managers, leaders and organizational executives with planning and formulating teams as well as improve communications from a clarity and perception standpoint. For instance, a team that has Indian Americans with Masters and Doctoral degrees can perceive innovation and outcomes to be new, improve or change as opposed to a team with members having bachelor's degrees may only be able to translate the vision and plan to fall under the categories of new or change/ improve, when it comes to perception of innovation.

The finding suggests that while selecting and managing a diverse team from a gender, generational and cultural standpoint, the understanding on perception of innovation and how members of the team define or categorize as innovation is crucial for the success of the team from a strategic execution standpoint. The incorrect mix of employees on a team with an uncommon understating of the goals or outcomes from a perception standpoint could lead to internal conflict and result in the team being ineffective. For instance, one of the findings from the factor analysis of the interactions between gender and functional association, suggested that males have a broader variation in their perception of innovation as opposed to females, who narrowly differentiated the three constructs of innovation and perceived them to be close to one another. Appropriate approach with an understanding of the culture background and definition or perception of innovation in homogenous groups could help improve diversity, ideation, creativity and execution of teams although the perceptions would start at the individual level. Thereby, resulting high performance teams or increased customers for a product or services standpoint, whose impact and benefits range holistically from a regional/ national to global dispersed and diverse environments.

Limitations

The research has several limitations that might affect the overall results. One of the first limitations is the researcher's experience level. As opposed to an experienced practitioner, the researcher, as a doctoral student did not bring much experience to this study. An experienced practitioner could anticipate some of the problems that the researched experienced and might have developed a different approach that would have yielded improved results and a simplified study. In addition, the study did not involve any sponsoring groups, however, the responses were 100% usable and the instrument proved to be reliable with Chronbach alpha coefficients of greater than 0.8, indicating good internal consistency. A greater sample might further enhance the reliability of the study. This survey was web-based and Qualtrics pre-qualified using a panel Indian-Americans, who in turn responded to demographic questions to indicate gender, age and functional association. With no incentives to misidentify, it may not be possible to verify this information on the section from web-surveys.

Recommendations

The Indian-American demography merits further research, especially with new factors such as education, which highlighted the dichotomy in the perception of innovation within the homogenous group. Further, extending the study into the Indian sub-continent and researching the definitions of perception could yield beneficial information and understanding of how Indian from various parts of the country in India perceive innovation as well. Further, Indian Americans in the United States are one of the fastest growing populations. India as a fast and rapidly evolving economy and Indian Americans as contributors to the economy in the United States, along with their existing ability of higher earnings, make the demography important from a team

building, sales, and product and service delivery perspective. An optimal and advanced understanding of perception of innovation and how innovation is seen and defined by Indian Americans, could benefit the demography, companies and the overall economy.

As the research and body of knowledge are broadened, additional dimensions such as education level, regional affiliation (within India), industry affiliation/ verticals, coupled with gender, age and functional association could prove beneficial from a management and sales perspective as applied to strategy and transformation of organization, products and services.

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APPENDIX A. STATEMENT OF ORIGINAL WORK

Academic Honesty Policy

Capella University's Academic Honesty Policy ([3.01.01](#)) holds learners accountable for the integrity of work they submit, which includes but is not limited to discussion postings, assignments, comprehensive exams, and the dissertation or capstone project.

Established in the Policy are the expectations for original work, rationale for the policy, definition of terms that pertain to academic honesty and original work, and disciplinary consequences of academic dishonesty. Also stated in the Policy is the expectation that learners will follow APA rules for citing another person's ideas or works.

The following standards for original work and definition of *plagiarism* are discussed in the Policy:

Learners are expected to be the sole authors of their work and to acknowledge the authorship of others' work through proper citation and reference. Use of another person's ideas, including another learner's, without proper reference or citation constitutes plagiarism and academic dishonesty and is prohibited conduct. (p. 1)

Plagiarism is one example of academic dishonesty. Plagiarism is presenting someone else's ideas or work as your own. Plagiarism also includes copying verbatim or rephrasing ideas without properly acknowledging the source by author, date, and publication medium. (p. 2)

Capella University's Research Misconduct Policy ([3.03.06](#)) holds learners accountable for research integrity. What constitutes research misconduct is discussed in the Policy:

Research misconduct includes but is not limited to falsification, fabrication, plagiarism, misappropriation, or other practices that seriously deviate from those that are commonly accepted within the academic community for proposing, conducting, or reviewing research, or in reporting research results. (p. 1)

Learners failing to abide by these policies are subject to consequences, including but not limited to dismissal or revocation of the degree.

Statement of Original Work and Signature

I have read, understood, and abided by Capella University's Academic Honesty Policy (3.01.01) and Research Misconduct Policy (3.03.06), including the Policy Statements, Rationale, and Definitions.

I attest that this dissertation or capstone project is my own work. Where I have used the ideas or words of others, I have paraphrased, summarized, or used direct quotes following the guidelines set forth in the *APA Publication Manual*.

Learner name

and date Chandragupta Gudena, March 3, 2014

Mentor name

and school Dr. Gregory McLaughlin, Capella University

APPENDIX B. SURVEY INSTRUMENT

INDIAN AMERICANS INNOVATION SURVEY

Background: The first seven questions are about your general background.

1 Please indicate your gender

- ☐ Male
- ☐ Female

2 What is your age?

- ☐ Born before 1947
- ☐ Born between 1947 and 1963
- ☐ Born between 1964 and 1979
- ☐ Born after 1980

3 In what country do you reside?

- ☐ United States
- ☐ Other (Please Specify) _____

4 What is your highest level of education?

- ☐ Did not complete High School
- ☐ High School or Equivalent
- ☐ Certificate
- ☐ Two-year degree
- ☐ Bachelor's degree
- ☐ Master's degree
- ☐ Doctoral degree

5 Which category best describes your position

- ☐ CEO/President
- ☐ Vice President
- ☐ Director
- ☐ Manager
- ☐ Supervisor
- ☐ Employee
- ☐ Other (please specify) _____

6 How many years in your present position? (Check only one box)

- ☐ Less than 1 year
- ☐ 1-3 years
- ☐ 4-6 years
- ☐ 7-10 years
- ☐ Over 10 years

7 Identify the type of Organizations where you worked

- ☐ Small business (less than 50 employees)
- ☐ Non-profit
- ☐ Governmental agency
- ☐ Education
- ☐ Consulting
- ☐ Manufacturing
- ☐ Healthcare
- ☐ Process industry
- ☐ Services
- ☐ Hospitality or Travel
- ☐ Legal
- ☐ Construction
- ☐ Food
- ☐ Agriculture
- ☐ Other
- ☐ Other (please specify) _____

8 Which of the following best describe your job function?

- ☐ Leadership
- ☐ Managerial
- ☐ Technical
- ☐ Engineering
- ☐ Financial
- ☐ Administrative
- ☐ Clerical
- ☐ HR
- ☐ Other (Please specify) _____

Environment Related: The following twelve statements asks you about your relation to your work environment.

9 Choose the best response that describes your present work environment

	Strongly disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
a. The work environment supports creativity	☐	☐	☐	☐	☐
b. I have confidence in my own abilities to solve problems	☐	☐	☐	☐	☐
c. My workplace provides me with challenges	☐	☐	☐	☐	☐
d. The workplace enables me to be creative	☐	☐	☐	☐	☐
e. The work environment is open to new ideas	☐	☐	☐	☐	☐
f. There is a sense of cooperation among employees	☐	☐	☐	☐	☐
g. Management rewards improvements	☐	☐	☐	☐	☐
h. Work demands are not overburdening	☐	☐	☐	☐	☐
i. My workplace encourages	☐	☐	☐	☐	☐

change j. Trust is a value in my workplace	☐	☐	☐	☐	☐
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Innovation Related: To different people, the concept of INNOVATION can mean different things. In the following ten questions, we would like to know what you think.

10 In your opinion, innovation can be defined as

	Strongly disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
a. Inventing something entirely new	☐	☐	☐	☐	☐
b. Generating new ideas	☐	☐	☐	☐	☐
c. Improving something that already exists	☐	☐	☐	☐	☐
d. Enhancing the desirability of something	☐	☐	☐	☐	☐
e. "Thinking out of the box"	☐	☐	☐	☐	☐
f. Taking something that already exists and making it better	☐	☐	☐	☐	☐
g. Changing the technology by making the previous version obsolete	☐	☐	☐	☐	☐
h. Adopting something that has been tried elsewhere	☐	☐	☐	☐	☐
i. Using a different perspective	☐	☐	☐	☐	☐

j. Introducing change to make something better	☐	☐	☐	☐	☐
--	-----------------------	-----------------------	-----------------------	-----------------------	-----------------------

11 As far as you are concerned, innovation can occur in a?

	Strongly disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
a. A product	☐	☐	☐	☐	☐
b. A service	☐	☐	☐	☐	☐
c. A technology	☐	☐	☐	☐	☐
d. A process	☐	☐	☐	☐	☐
e. A person	☐	☐	☐	☐	☐

12 Who benefits most from an innovation?

	Strongly disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
a. The individual or group responsible for the innovation	☐	☐	☐	☐	☐
b. The employer	☐	☐	☐	☐	☐
c. The industry	☐	☐	☐	☐	☐
d. The society	☐	☐	☐	☐	☐

13 Once an innovation is identified, what are the next steps? (Please rank the following statements in order of importance from 1 to 6)

- _____ a. It should be thoroughly evaluated
- _____ b. An implementation plan initiated
- _____ c. A cost-benefit analysis completed
- _____ d. An assessment of its short-term and long-term potential
- _____ e. Determine how it advances, improves, or changes performance
- _____ f. Determine if it meets its stated objectives

14 The value of an innovation should be judged upon:

	No Value	Little value	Has some value	Is valued	Of great value
a. How new or novel it is	☐	☐	☐	☐	☐
b. how many people it involves	☐	☐	☐	☐	☐
c. How much it costs	☐	☐	☐	☐	☐
d. How long it takes	☐	☐	☐	☐	☐
e. How much improvement does it make	☐	☐	☐	☐	☐
f. How well the technology is advanced	☐	☐	☐	☐	☐
g. The extent of change that occurs	☐	☐	☐	☐	☐

15 Check the box that best matches your agreement (disagreement) with each "value" statement

	Strongly agree	Agree	Neither Agree nor Disagree	Disagree	Strongly Disagree
a. The core mission of my company has my support	☐	☐	☐	☐	☐
b. The strengths of my organization help me to accomplish my work	☐	☐	☐	☐	☐
c. Weaknesses in the organization do not deter my enthusiasm	☐	☐	☐	☐	☐
d. Individuals are committed to the success of the organization	☐	☐	☐	☐	☐
e. Management effectively motivates employees	☐	☐	☐	☐	☐
f. Applying positive values to the work environment is helpful	☐	☐	☐	☐	☐
g. Accepting the assigned priorities for the most critical tasks	☐	☐	☐	☐	☐

is easy					
h. My beliefs can conform to achieve the desired objective	☐	☐	☐	☐	☐
i. Empowering others is a value that generates success	☐	☐	☐	☐	☐
j. Teams can achieve greater success than individuals	☐	☐	☐	☐	☐

16 For your organization, rank in the order of importance, whom you think is in a better position to initiate an innovation?

- _____ The board of directors
- _____ The marketing personnel
- _____ The quality staff
- _____ Accounting and finance
- _____ The purchasing people
- _____ Experts (consultants)
- _____ HR (human resources)

17 Organizations sometimes embark on innovative activities Choose the single most critical reason.

- ☐ a. Because they want to increase the bottom line
- ☐ b. Because their competitors are innovative
- ☐ c. Because they want to do better than their competitors
- ☐ d. Because of pressure from accreditors and regulators
- ☐ e. Because they happen to have creative personnel
- ☐ f. In an attempt to reduce costs
- ☐ g. Since they have the financial resources
- ☐ h. Due to decreased reimbursement

18 The key to one's ability to innovate is to (Choose only one response)

- ☐ a. Be born with the right talent
- ☐ b. Be able to think creatively
- ☐ c. Do a postgraduate course
- ☐ d. Have extensive experience in the field
- ☐ e. Possess good interpersonal skills
- ☐ f. Be good at putting theory into practice
- ☐ g. Other (please specify) _____

19 Which of the following organizational factors is paramount to realizing one's innovation potential? (Choose only one response)

- ☐ a. Regular employee performance appraisals
- ☐ b. Freedom to work on areas of greatest interest
- ☐ c. Feedback from superiors
- ☐ d. Recognition and appreciation
- ☐ e. Contacts with like-minded colleagues
- ☐ f. Training and education
- ☐ g. Encouragement to take risks
- ☐ h. Monetary rewards
- ☐ i. Tolerance of nonconformity
- ☐ j. Opportunity to work alone rather than on a team

20 Which of the following organizational factors do you think is most emphasized in the organization where you work (choose only one response)

- ☐ a. Regular employee performance appraisals
- ☐ b. Freedom to work on areas of greatest interest
- ☐ c. Feedback from superiors
- ☐ d. Recognition and appreciation
- ☐ e. Contacts with like-minded colleagues
- ☐ f. Creativity training programs
- ☐ g. Encouragement to take risks
- ☐ h. Monetary rewards
- ☐ i. Tolerance of nonconformity
- ☐ j. Opportunity to work alone rather than on a team

APPENDIX C. FACTOR ANALYSIS OVERALL

Table C1

Factor Analysis of all 200 participants at 95% confidence level

Level	N	Mean	StDev.
C30 (New)	200	4.2450	0.6130
C31 (Improve+Change)	200	4.0191	0.6779

“New” scores higher than “Improve or Change”. The component C30 represents the “new” construct and C31 represents “Improve or change”. Respondents did not seem to differentiate between Improve and Change.

Table C2

Descriptive Statistics

	Mean	Std. Deviation	Analysis N
In your opinion, innovation can be defined as-a. Inventing something entirely New	4.16	.833	200
In your opinion, innovation can be defined as-b. Generating new ideas	4.30	.702	200
In your opinion, innovation can be defined as-c. Improving something that already exists	4.03	.905	200

In your opinion, innovation can be defined as-d. Enhancing the desirability of something	4.07	.777	200
In your opinion, innovation can be defined as-e. "Thinking out of the box"	4.28	.809	200
In your opinion, innovation can be defined as-f. Taking something that already exists and making it better	4.00	.894	200
In your opinion, innovation can be defined as-g. Changing the technology by making the previous version obsolete	3.93	.916	200
In your opinion, innovation can be defined as-h. Adopting something that has been tried elsewhere	3.85	1.016	200
In your opinion, innovation can be defined as-i. Using a different perspective	4.12	.754	200
In your opinion, innovation can be defined as-j. Introducing change to make something better	4.14	.806	200

Table C3

KMP and Bartlett's Tests

Kaiser-Meyer-Olkin Measure of Sampling Adequacy	.867
Bartlett's Test of Sphericity Approx. Chi-Square	945.012
Df	45
Sig.	.000

Table C4

Communalities

	Initial	Extraction
In your opinion, innovation can be defined as-a. Inventing something entirely New	1.000	.666
In your opinion, innovation can be defined as-b. Generating new ideas	1.000	.717
In your opinion, innovation can be defined as-c. Improving something that already exists	1.000	.662
In your opinion, innovation can be defined as-d. Enhancing the desirability of something	1.000	.628
In your opinion, innovation can be defined as-e. "Thinking out of the box"	1.000	.421
In your opinion, innovation can be	1.000	.687

defined as-f. Taking something that already exists and making it better		
In your opinion, innovation can be defined as-g. Changing the technology by making the previous version obsolete	1.000	.512
In your opinion, innovation can be defined as-h. Adopting something that has been tried elsewhere	1.000	.705
In your opinion, innovation can be defined as-i. Using a different perspective	1.000	.630
In your opinion, innovation can be defined as-j. Introducing change to make something better	1.000	.612

Extraction Method: Principal Component Analysis

Table C5

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	4.975	47.754	49.754	4.975	49.754	49.754
2	1.264	12.642	62.396	1.264	12.642	62.396
3	.805	8.046	70.442			

4	.649	6.491	76.933
5	.589	5.894	82.827
6	.463	4.628	87.455
7	.436	4.356	91.811
8	.300	2.995	94.806
9	.282	2.820	97.627
10	.237	2.373	100.00

Table C6

Principal Component Analysis

Component	Rotation Sum of Squared Loadings		
	Total	% of Variance	Cumulative %
1	3.901	39.013	39.013
2	2.338	23.384	62.396
3			
4			
5			
6			
7			
8			
9			
10			

Extraction Method: Principal Component Analysis

Table C7

Component Matrix

	Component	
	1	2
In your opinion, innovation can be defined as-a. Inventing something entirely	.522	.627
New In your opinion, innovation can be defined as-b. Generating new ideas	.610	.587
In your opinion, innovation can be defined as-c. Improving something that already exists	.763	-.283
In your opinion, innovation can be defined as-d. Enhancing the desirability of something	.754	-.272
In your opinion, innovation can be defined as-e. "Thinking out of the box"	.522	.385
In your opinion, innovation can be defined as-f. Taking	.768	-.310

something that already exists and making it better	.710	-.088
In your opinion, innovation can be defined as-g. Changing the technology by making the previous version obsolete	.771	-.332
In your opinion, innovation can be defined as-h. Adopting something that has been tried elsewhere	.793	.023
In your opinion, innovation can be defined as-i. Using a different perspective	.777	.091
In your opinion, innovation can be defined as-j. Introducing change to make something better		

Extraction Method: Principal Component Analysis

2 Components extracted.

Table C8

Rotated Component Matrix

	Component	
	1	2

In your opinion, innovation can be defined as-a. Inventing something entirely	.102	.810
New In your opinion, innovation can be defined as-b. Generating new ideas	.198	.823
In your opinion, innovation can be defined as-c. Improving something that already exists	.795	.171
In your opinion, innovation can be defined as-d. Enhancing the desirability of something	.774	.172
In your opinion, innovation can be defined as-e. "Thinking out of the box"	.233	.606
In your opinion, innovation can be defined as-f. Taking something that already exists and making it better	.815	.152
In your opinion, innovation can be defined as-g. Changing the technology by making the previous version obsolete	.646	.308

In your opinion, innovation can be defined as-h. Adopting something that has been tried elsewhere	.829	.135
In your opinion, innovation can be defined as-i. Using a different perspective	.657	.446
In your opinion, innovation can be defined as-j. Introducing change to make something better	.606	.495

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

a. Rotation converged in 3 iterations.

Table C9

Component Transformation Matrix

Component	1	2
1	.843	-.538
2	-.538	.843

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

Table C10

Reliability Statistics

	N	%
Valid	200	100.0
Excluded	0	0.0
Total	200	100.0

Table C11

Cronbach Alpha

Cronbach's Alpha	N of items
0.883	10